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A novel explainable AI framework for medical image classification integrating statistical, visual, and rule-based methods

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ABSTRACT

Artificial intelligence and deep learning are powerful tools for extracting knowledge from large datasets, particularly in healthcare. However, their black-box nature raises interpretability concerns, especially in high-stakes applications. Existing eXplainable Artificial Intelligence methods often focus solely on visualization or rule-based explanations, limiting interpretability's depth and clarity. This work proposes a novel explainable AI method specifically designed for medical image analysis, integrating statistical, visual, and rule-based explanations to improve transparency in deep learning models. Statistical features are derived from deep features extracted using a custom Mobilenetv2 model. A two-step feature selection method – zero-based filtering with mutual importance selection – ranks and refines these features. Decision tree and RuleFit models are employed to classify data and extract human-readable rules. Additionally, a novel statistical feature map overlay visualization generates heatmap-like representations of three key statistical measures (mean, skewness, and entropy), providing both localized and quantifiable visual explanations of model decisions. The proposed method has been validated on five medical imaging datasets – COVID-19 radiography, ultrasound breast cancer, brain tumor magnetic resonance imaging, lung and colon cancer histopathological, and glaucoma images – with results confirmed by medical experts, demonstrating its effectiveness in enhancing interpretability for medical image classification tasks.

1. Introduction

Artificial Intelligence (AI) and Deep Learning (DL) have revolutionized domains requiring high predictive accuracy, particularly in high-stakes applications like healthcare, criminal justice, and beyond (Aamir et al., 2021) in recent years. In healthcare, DL models have significantly improved diagnostic accuracy across various imaging modalities, including Magnetic Resonance Imaging (MRI), chest radiographs, histopathology, and ultrasounds (Qayyum et al., 2024; Ullah et al., 2023). Despite their success, these models face criticism for their lack of transparency and interpretability, raising trust and ethical concerns, especially in critical fields like healthcare (Rudin, 2019; Poon and Sung, 2021).

To address these challenges, eXplainable Artificial Intelligence (XAI) and Interpretable Artificial Intelligence (IAI) have emerged as solutions (Ullah et al., 2024a). XAI techniques explain the decision-making logic of DL models, enhancing transparency and trust without sacrificing accuracy. Post hoc XAI methods, such as feature importance rankings (Wojtas and Chen, 2020), saliency maps (Gomez et al., 2022), and gradient-based visualizations (Selvaraju et al., 2017), aim to provide insights into black-box models. In contrast, IAI develops inherently transparent models, such as decision trees or rule-based systems, to present decision-making processes in a human-readable format (Reyes et al., 2020; De Falco et al., 2023).

However, some key limitations persist. While effective for classification, deep features extracted from convolutional layers are abstract and lack semantic interoperability for experts, making it difficult to understand how specific features influence decisions. Additionally, most visualization methods, like saliency maps, only highlight general input regions relevant to predictions but fail to provide detailed feature-level explanations. Rule-based approaches often lack integration with visual interpretations, leading to fragmented insights. This disjointedness is

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particularly problematic in healthcare, where transparent and precise explanations are crucial for trust and decision-making.

To address these issues, this work proposes statistical, visual, and rule-based explainable (SVIS-RULEX). This new framework integrates statistical, visual, and rule-based explanations to explain how DL-based models make decisions when applied to medical images. Unlike traditional methods, this framework transforms deep features into interpretable statistical attributes, enabling traceable and quantifiable predictions. Moreover, a novel method called SFMOV is introduced. SFMOV enhances visualization by overlaying statistical measures (e.g., mean, skewness, entropy) onto feature maps, providing localized and feature-specific insights. By merging statistical, visual, and rule-based explanations, this approach ensures greater transparency and interpretability, which is crucial for healthcare applications.

To validate its robustness and generalizability, the SVIS-RULEX framework is evaluated on five diverse medical imaging datasets from Kaggle, covering chest radiographs, breast ultrasounds, brain MRI, histopathological and retinal fundus images. This broad evaluation demonstrates the framework's adaptability and effectiveness across various imaging modalities and medical conditions.

The major contributions of this work are summarized as follows:

1. A dual-path approach integrating visual and rule-based explanations for enhanced interpretability while maintaining classification performance.
2. Deep feature extraction using a custom MobileNetv2 model, capturing high-dimensional representations from diverse medical imaging modalities.
3. Introduction of a novel statistical feature engineering approach, generating 26 human-readable attributes (e.g., mean, variance, skewness, entropy) from deep features for improved interpretability.
4. Development of the novel ZFMIS strategy for feature ranking and selection, ensuring compact and informative feature sets.
5. Rule-based interpretation using DT and RuleFit, providing human-understandable insights into the model's decision-making process.
6. Proposal of the novel SFMOV technique to visualize statistical overlays (mean, skewness, entropy) of convolutional features, weighted by dense-layer properties, for deeper interpretability.
7. Clinical validation of the SFMOV visualizations with radiologists, confirming their diagnostic relevance and practical significance.

The remainder of the paper is structured as follows: Section 2 reviews related work. Section 3 details the proposed methodology. Section 4 presents and discusses the results, while Section 5 concludes the paper.

2. Related work

Recently, AI has proved extremely useful in the healthcare domain, specifically in medical image analysis (Van der Velden et al., 2022; Bevilacqua et al., 2024). In literature, DL models are largely used for image classification tasks in the healthcare domain because of their high classification performance. This high classification performance has often been achieved through increased model complexity. These complex models are usually called back box systems because their decisions are hard to understand and interpret. Hence, these models are difficult to be trusted, which restricts the use of these models in critical sectors, such as healthcare. However, XAI has emerged to address the need for transparency in complex black box classification models.

In the literature, the two main methods for achieving explainability in Machine Learning (ML) are post-hoc explainability and ante-hoc explainability. The kind of insights that these approaches may provide, as well as how and when interpretability is provided, vary widely. Post-hoc XAI techniques are applied to analyze the model after training the model whereas in the case of ante-hoc the models are intrinsically explainable (Retzlaff et al., 2024). Traditional ML models, such as DT, linear or logistic regression (Tripepi et al., 2008), Bayesian models (Wang et al., 2017), rule-based (Nauck, 2003), and k-nearest neighbor models (Papernot and McDaniel, 2018) are inherently interpretable (glass-box models) whereas DL models are not. These models are more transparent and explainable from the start and incorporate interpretability directly into the model (Sarkar et al., 2022). Linear interpretable models offer direct insight into feature importance through their coefficients, allowing users to understand feature influence and interaction in a straightforward, step-by-step manner. DTs are also the most prominent interpretable ML models. DTs split the data multiple times to produce separate subsets of the dataset or features based on certain threshold values in the features, with each instance belonging to one subset (Safavian and Landgrebe, 1991). The tree structure of these models starting from the root node to the successive nodes and edges until you arrive at the output (leaf node) makes these models interpretable. Each path from the root node to the leaf can be represented as a hierarchical structure of if-then-else-rules that are straightforward for humans to grasp and comprehend (Christoph, 2020).

Although in-model XAI methods promote transparency, it often comes at the expense of model performance and complexity. This is because interpretable models such as DTs or linear models struggle to capture complex patterns in high-dimensional data. Whereas DL models capture more complex patterns, but they are black boxes and lack interpretability. To address this, DT have been widely employed as surrogate models in a post hoc manner, providing an interpretable approximation of the decision boundaries of black boxes. In Confalonieri et al. (2021), Trepan Reloaded was developed based Trepan algorithm that extracts surrogate decision trees from black-box models, incorporating ontologies to improve the comprehensibility of the explanations.

In this paper, the focus is on adding explainability post hoc to existing DL models. Post-hoc XAI techniques interpret, analyze, and increase the interpretability of trained black box models providing insights into how the model makes specific predictions. The advantages of the post hoc XAI approaches include providing explanations without modifying the architecture and applicability to various black box models. In the medical image analysis domain, post hoc explainability methods are most widely used nowadays.

In post-hoc explainability, a white box surrogate model is constructed for the trained black box model (Adadi and Berrada, 2018). Post-hoc techniques transfer the learned knowledge from a complex black-box model into a simple white-box or surrogate model (Confalonieri et al., 2021). These are independent of models and are applied once the trained model has made the prediction. These techniques provide an approximate explanation of what the black-box model is doing, but not exactly how or why it arrived at a prediction. This approximate explanation provides a close and useful understanding of the black-box model's internal logic and does not negatively impact the underlying model's predictive accuracy (Burkart and Huber, 2021).

Post-hoc explainability techniques traditionally generate two types of interpretability i.e., global and local interpretability (Stiglic et al., 2020). Global interpretability explains the overall logic and behavior behind all possible outcomes of a black box model (Nian et al., 2024). This helps to identify the most important rules and patterns learned during model training and display them through the model's structure and parameters. The coefficients of a linear regression model and the rules formed along the paths from root nodes to leaf nodes of a decision tree are examples of global interpretability (Stiglic et al., 2020). On the other hand, local interpretability explains a single prediction and the impact of each feature on a specific outcome. This granular approach is often more accurate than global interpretability because instead of explaining the entire model and its overall logic, it examines every aspect of the model's decision-making process.

The two most widely used techniques for local interpretability in ML are SHapley Additive exPlanations (SHAP) (Lundberg, 2017)

and Locally Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016a). SHAP is based on cooperative game theory, specifically Shapley values, which assign an importance value to each feature in a prediction. This method provides a correct and detailed explanation of the influence of each feature on the model's output in a given instance. In contrast, LIME generates a simple surrogate model, such as linear regression, to replicate the behavior of the difficult model in the local neighborhood of the instance under examination. Both LIME and SHAP focus on explaining individual predictions rather than overall model behavior, making them suited for applications requiring instance-level insights.

Post-hoc explainability can be applied in two ways i.e., model-specific and model-agnostic interpretability. The model-specific interpretability techniques provide explanations about the internal representation or learning process of a particular model. Therefore, these techniques are inherently tailored for every model and cannot be used to explain any other model. Whereas model-agnostic explanations approximate the underlying behavior of models and generate end-user explanations independent of the internal logic of models (Ribeiro et al., 2016b). Model-agnostic methods have the potential to be applied to any black box model because they do not consider the internal components of a model, such as weights or structural information, and are not model-specific explainers. Therefore, they have wide-scale applications. Post-hoc model-agnostic explanations fall into the broader category of explaining by removing (Covert et al., 2021). To identify feature importance these methods remove input features and examine their impact on the prediction. Features are considered important only if their removal significant impact on prediction (Troncoso-García et al., 2023a).

Kim et al. (2018) proposed Testing with Concept Activation Vectors (TCAV), a post-hoc explainability method that interprets model predictions in terms of human-interpretable concepts rather than low-level features. Unlike saliency maps that report feature importance at the pixel or input level, TCAV estimates a model's dependency on high-level, user-specified concepts by computing directional derivatives along concept vectors in neural network activations. This technique provides an overall numerical measure of a concept's salience to the decision of a model and is, therefore, particularly useful in situations where model thinking is critical, such as healthcare. The avoidance of the requirement for retraining the model provided by TCAV renders it an excellent tool to use in understanding biases and dependencies on features for deep learning models. Given our research focus on integrating statistical, rule-based, and visual explanations, TCAV can be regarded as a complementary perspective that emphasizes concept-level interpretability.

Furthermore, post-hoc methods can be categorized into several key types based on the way they convey interpretability to the user, such as rule-based interpretability, visualization-based techniques, feature importance analysis, and example-based methods. These approaches aim to shed light on complex, "black-box" models while preserving their predictive accuracy.

Rule-based methods provide human-readable explanations by extracting logical statements or interpretable rules. These approaches are mainly used for structured problems with tabular data. DTs and rule extraction algorithms such as TREPAN (Confalonieri et al., 2021) are examples of rule-based interpretability (Troncoso-García et al., 2023b). These methods explain the behavior of complex models using a simpler surrogate model, such as a DT, that represents decision boundaries in terms of "if-then-else" rules. The explanations in the form of rules help non-technical stakeholders to understand the reasoning behind predictions.

Visualization techniques are the most commonly used post-hoc explainability methods. These methods use visual explanations to highlight the contribution of input features to a model's predictions. Recently, Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) is most commonly used in image-based tasks to identify the most important regions of an image that influence a particular prediction. This method overlays the heatmaps generated using gradient information from the trained model on the input image to highlight significant areas. In the case of textual data, attention maps or saliency visualizations help users understand which words contributed to a decision. Visualization methods are particularly effective in domains like medical imaging and computer vision, where they enable domain experts to validate model decisions intuitively.

Feature importance approaches are model-agnostic techniques that measure the contribution of each input feature to a model's prediction. Methods like SHAP and LIME explain individual predictions by perturbing the input features and assessing the changes in model output. These approaches produce global feature rankings or local explanations, allowing users to determine the most relevant aspects for a single prediction or overall model behavior.

Example-based interpretability is founded on case-based reasoning, in which explanations are offered by referring to specific examples from the training dataset (Poché et al., 2023). This is consistent with the psychological mechanics of human thinking, making example-based explanations natural and easy for consumers to grasp. Indeed, people learn and reason by creating mental models of concepts based on instances. This category includes techniques, such as counterfactual explanations (Verma et al., 2020), that reveal the minimal adjustments to the input features required to modify the prediction outcome, providing consumers with actionable insights. Another example is prototype-based (Casalino et al., 2024) approaches that choose representative instances from the training data that are comparable to the input instance, allowing users to better understand model behavior through analogous scenarios.

Evaluating explainability approaches is a subjective and challenging task since interpretability often depends on the specific application domain and the target audience's degree of knowledge. While ante-hoc techniques include interpretability directly into the model's structure, post-hoc approaches rely on additional tools to extract insights from complex models. The effectiveness of these tactics depends on end users' capacity to comprehend the reasoning behind predictions and confidently act upon them. Initiatives like user-centered validation and standardized evaluation frameworks are essential for assessing the suitability of XAI approaches across a range of domains as research advances. The effectiveness of an explainable model ultimately depends on its ability to bridge the gap between the complex technicalities of black box models and the real-world needs of its users, encouraging trust and informed decision-making.

Existing explainability methods for black-box models face several limitations. Post-hoc explainability approaches in the literature focus solely on one aspect, visualization, rule-based explanation, or statistical analysis, without integrating them to provide a holistic understanding of model behavior. DTs are used for rule-based interpretability but are not integrated with other explainability techniques (e.g., statistical and visual methods). Additionally, post-hoc methods, mostly rely on gradient-based attributions or raw feature importance and often overlook the potential of statistical transformations derived from deep features, which can offer richer insights into data patterns. Techniques such as TCAV provide concept-based explanations but require predefined human-interpretable concepts, limiting their applicability in scenarios where defining such concepts is challenging. Furthermore, visual explanation techniques, neglect the visualization of statistical features critical for classification and focus only on activation or feature attribution maps.

Given the existing gaps in the literature, such as the lack of integrated statistical, visual, and rule-based approaches for explainability, and the limitations of conventional visualization methods like generic saliency maps, this work proposes the SVIS-RULEX approach. This novel method addresses these issues by combining statistical, visual, and rule-based explainability techniques for a more comprehensive

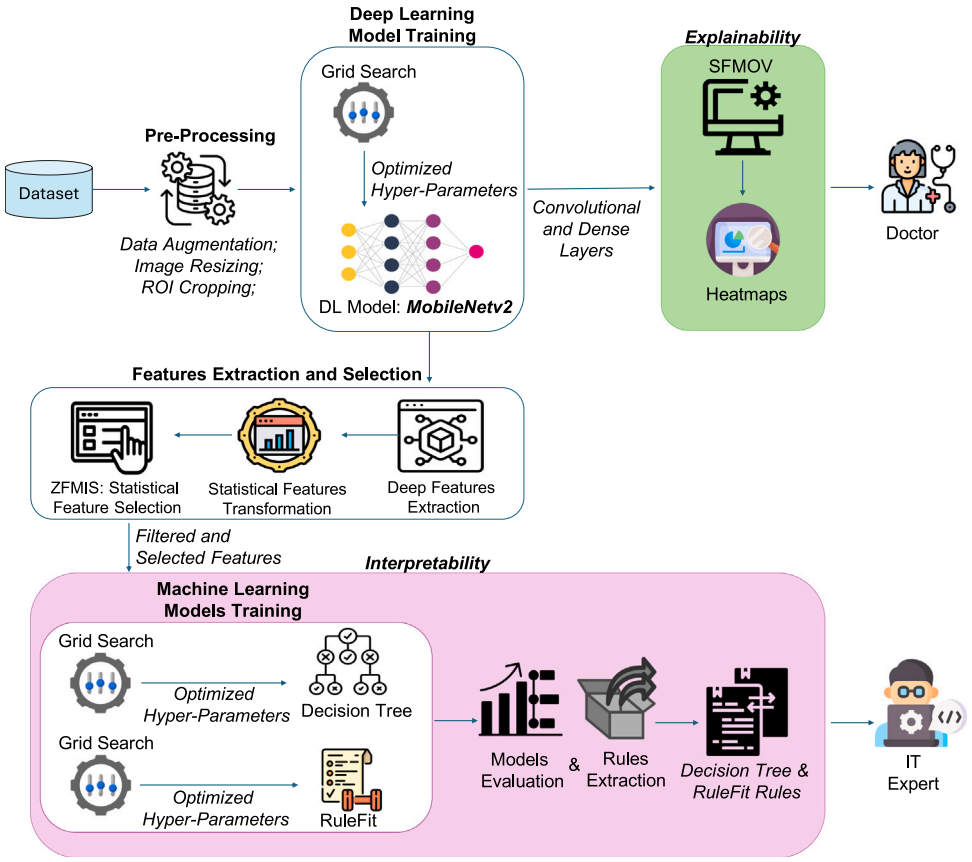


Fig. 1. Workflow of the proposed approach.

post hoc interpretation of deep learning models. Moreover, the proposed SFMOV goes beyond traditional heatmaps by providing localized and quantifiable insights into model decisions, using key statistical measures (mean, skewness, and entropy) to enhance transparency and interpretability.

3. Methodology

This work integrates rule-based modeling and visualizations with statistical feature engineering and DL-based feature extraction to produce human-understandable explanations. This method bridges the gap between interpretable techniques' transparency and black-box DL models' predictive power. Fig. 1 shows an overview of the proposed method. First, some existing techniques are used in preprocessing, like data augmentations, image resizing, and region of interest cropping. The resultant dataset is then fed into the custom MobileNetv2 DL model for feature extraction to capture complex visual patterns and get more discriminative and informative deep features, which are crucial for accurate classification. These extracted features are then used for statistical transformation to compute interpretable, summary-level insights. Next, a novel ZFMIS technique ranks and filters features based on their interpretive value. The filtered and selected features are then used to train the decision tree and RuleFit interpretable rule-based models. Finally, Grad-CAM-inspired heatmaps (SFMOV) are generated to visualize the most important statistical features and highlight spatial regions of the images that contribute significantly to the model's decision-making process. Visual overlays enhance statistical interpretation by illustrating how specific patterns affect categorization judgments, providing stakeholders with an intuitive understanding of the prediction model's behavior, a model-agnostic visual explanation technique.

The pseudocode of the proposed methodology is presented in Algorithm 1. Each step is discussed in detail in the subsequent sections.

3.1. Pre-processing

In this stage, the dataset's images are preprocessed to meet the image input requirements of the custom MobileNetv2 model, utilizing existing image preparation and enhancement methods. The preprocessing in this work is variable to enhance the efficacy of feature extraction and model performance, and various techniques may be used depending on the dataset and imaging modalities. The following methods are mainly used in this study.

Region of Interest (ROI) cropping. Region of Interest (ROI)-based cropping is used in the COVID-19 Radiography Database to isolate the important chest regions of the images according to the segmentation mask and remove the background parts. This is crucial for improving the quality of input data and the relevance of the features fed into the model while lowering computing overhead, as only the most interesting areas were examined.

$$I_c = I_o[x_{min} : x_{max}, y_{min} : y_{max}] \tag{1}$$

Where I_c stands for cropped image, I_o stands for the original image, (x_{min}, x_{max}) and (y_{min}, y_{max}) stands for the coordinates of the bounding box enclosing the non-zero values in the segmentation mask (Minimum values represents start of mask whereas the maximum values represent end of mask).

Data augmentation. To reduce overfitting and enhance the diversity of the training dataset the dataset size is artificially expanded through a series of image transformations using the Albumentations library (Buslaev et al., 2020). A combination of photometric, geometric, and simulated atmospheric transformations is included to imitate real-world variations present in medical imaging.

Algorithm 1: Algorithm for deep feature extraction, statistical feature analysis, and rule-based classification with visual interpretation

Input: Input Image Dataset $\mathcal{ID} = \{I_1, I_2, \ldots, I_m\}$, Pre-trained Custom MobileNetV2 Model $\mathcal{CMM}$.
Output: Extracted Rules and statistical feature map overlay visualization (SFMOV).
Step 1: Deep Feature Extraction

- For each $I_x \in \mathcal{ID}$:
  - Extract deep feature vector $\mathbf{F}_{\text{deep}}(I_x) \in \mathbb{R}^d$ using $\mathcal{CMM}$.
- Construct deep feature matrix $\mathbf{F}_{\text{deep}} = [\mathbf{F}_{\text{deep}}(I_1), \mathbf{F}_{\text{deep}}(I_2), \ldots, \mathbf{F}_{\text{deep}}(I_m)]^T$.

Step 2: Statistical Feature Computation

- For each feature vector $\mathbf{F}_{\text{deep}}(I_x)$:
  - Compute statistical features:

$$\mathbf{F}_{\text{stat}}(I_x) = \left[mean(\mathbf{F}_{\text{deep}}(I_x)), variance(\mathbf{F}_{\text{deep}}(I_x)), \text{skew}(\mathbf{F}_{\text{deep}}(I_x)), entropy(\mathbf{F}_{\text{deep}}(I_j)), \ldots\right]$$

- Construct statistical feature matrix $\mathbf{F}_{\text{stat}} \in \mathbb{R}^{N\times 26}$.

Step 3: Feature selection

- Apply a two-stage feature selection algorithm for feature selection (Algorithm 2)

Step 4: Rule Extraction using Decision Tree and RuleFit

- Train decision tree classifier $\mathcal{DT}$ on $\mathbf{F}_{\text{selected}}(k)$ and extract decision rules $\mathcal{DTR}$.
- Train RuleFit model to extract additional rule sets $\mathcal{RFR}$.

Step 5: Visualization of Important Features

- Use mean, skewness, entropy, and a combined approach for visualization (Algorithm 3)

For each original image I_o, we have applied six types of augmentations

$$Transform(I_o) = I_r, I_f, I_{bc}, I_s, I_{rgbs}, I_{rf} \tag{2}$$

The I_r stands for rotation of ±5 degrees, I_f stands for a random flipping along both axes with a probability of 0.5, I_{bc} represents changes in random brightness and contrast of the images, I_s slight image shifts, I_{rgbs} shows a minor RGB shifts with a limit of ±10 units for each color channel, and I_{rf} denotes a random fog effect with a probability of 0.2.

Dataset partitioning. This study's datasets are split into training, validation, and testing sets to improve the performance of deep feature extraction and the effectiveness of statistical model classification. Each subject's images are exclusively assigned to the training, validation, or testing set, so no image from the same subject appears in more than one set. For deep feature extraction using a custom MobileNetV2 model, 70% of the data is used for training, while the remaining 30% is used for validation (15%) and testing (15%). For the classification of statistical features (derived from deep features) using decision tree and RuleFit, the training and validation set is combined into a single 85% training set, whereas, the remaining 15% of the data is used for testing purposes. The combined training and validation set is used to enhance the training data volume, generalizability, robustness, statistical stability, and classification performance of DT and RuleFit.

Image resizing. After augmentation and ROI cropping, the input images are resized to 224 × 224 according to the custom MobileNetv2 model's input image requirements. Hence the final preprocessed images can be obtained as

$$I_{preprocessed} = Resize(Augment(Crop(I_{original})), (224, 224)) \tag{3}$$

So, the cropping, augmentation, and resizing are applied sequentially to all dataset images.

3.1.1. Custom MobileNetv2 and deep feature extraction

DL models, including custom and pre-trained architectures, have been extensively used for feature extraction, enabling hierarchical pattern recognition in applications like COVID-19 detection, skin cancer classification, and brain tumor diagnosis (Ullah and Javed, 2021; Ghosh et al., 2024; Raza et al., 2024; Verde et al., 2023; Ullah et al., 2024b; Verde et al., 2021). Since this work proposes a general methodology, any DL model can be used for deep feature extraction. However, MobileNetV2 is used for its lightweight architecture, memory efficiency, and suitability for mobile and resource-constrained environments. MobileNetV2 employs depth-wise separable convolutions, inverted residuals, and linear bottleneck structures to reduce computational complexity while maintaining high representational power. Furthermore, the selection of MobileNetV2 is based on its superior classification performance compared to other DL models including DenseNet201 (Huang et al., 2017) and ResNet18 (He et al., 2016) on the studied dataset. This flexibility allows the proposed methodology to be adapted to other datasets and DL models, ensuring broad applicability across diverse domains such as chest radiographs, CT scans, MRI images, etc.

To adapt the MobileNetV2 model to this specific medical imaging task in this study, the original classification head is removed and replaced by a custom classification head. Before classification, the high-dimensional feature maps of the convolutional layers must be converted to a compact fixed-size feature vector. For this purpose, Global Average Pooling (GAP) is employed. GAP reduces each feature map's spatial dimensions by averaging all of its spatial values, essentially condensing each map into one representative value. This technique not only reduces the number of parameters and minimizes the risk of overfitting but also retains the most valuable global information from each feature map. The resulting fixed-size feature vector holds the pertinent features of the input image and is fed into both the deep statistical feature extraction and the custom classification head. To enhance model adaption, a fine-tuning strategy is utilized: all layers in MobileNetV2 are

frozen except the last 50, which are modified during training to learn target dataset-specific patterns while maintaining general knowledge from the pre-trained model.

Furthermore, a custom classification head with three dense (with ReLU activation functions), two batch normalization, and two dropout layers is used to classify the new data. The ReLU activation function is used to mitigate the vanishing gradient problem, batch normalization is applied to stabilize the learning process, and dropout layers are used to reduce overfitting. Finally, the softmax activation function is used to produce class probabilities.

The features are extracted from the fourth-last layer (second dense layer) of the custom MobileNetv2 model. The dense layer is chosen because it contains a high-dimensional feature map and retains both spatial information and an abstract representation of the images. Before feature extraction, the input images are processed and organized into training, validation, and test datasets. The data generator is used to supply a batch of input images to the model and extract a single feature vector with a corresponding label for each image ($Features = MobileNetV2(Image)$). The extracted features are saved in a format (pickle file) so that they can be retrieved and used in the future for classification and rule extraction.

3.1.2. Statistical feature engineering

Statistical features are considered as a consistent pattern of information or structure in raw data. In literature, various methods are used to extract statistical features including transform-based, statistical, and structural approaches. In Latif et al. (2018), statistical-based feature extraction methods were employed to extract first-order statistical features (entropy, kurtosis, variance, mean, skewness, and energy) and second-order statistical features (maximum probability, absolute value, angular second-moment energy, inertia, correlation, inverse difference, entropy) from magnetic resonance imaging (MRI) images. Such statistical features have proven valuable for enhancing medical image classification tasks, where ensemble approaches that combine deep spatial features with handmade statistical features have been widely used (Iqbal et al., 2024; Lai and Deng, 2018).

Building on this idea, statistical feature engineering complements DL by transforming the abstract, highdimensional representations extracted from DL models – such as MobileNetV2 – into interpretable statistical measures. Although deep features capture complex patterns and abstract representations, they often lack the semantic interpretability required for practical applications. However, statistical features, on the other hand, may not be inherently interpretable in isolation; they serve as structured and quantifiable descriptors of activation patterns, allowing for a structured understanding of feature distributions and help to bridge the gap between raw deep features and human-interpretable insights. Such statistical summaries provide a more tangible interpretation of how the network processes input data, making them useful for analyzing and explaining model decisions.

To address the limitation of deep features (lack of the interpretability required for practical applications), a comprehensive set of 26 statistical metrics is employed to provide a deeper analysis of the feature space. These metrics, including measures of central tendency, dispersion, and shape, provide a systematic way to highlight feature distributions within activation maps, thereby enhancing transparency in model decision-making. Additionally, statistical feature extraction allows for dimensionality reduction while preserving key characteristics, facilitating the training of interpretable models such as decision trees and RuleFit, which provide human-readable rules. The formulas and definitions of these statistical measures are detailed in Table 1. In the table, the symbol x denotes a particular feature value within a feature vector, and n denotes the total number of values in the feature vector.

Table 1
Statistical feature descriptions and formulas.

Statistical feature	Formula
Mean (μ)	$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
Median	Median of feature vector x
Variance (σ^2)	$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$
Geometric Mean	$\text{GM} = \exp(\frac{1}{n}\sum \ln(x_i))$
Standard Deviation (σ)	$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$
Minimum Value	$\min(x)$
Lag-1 Autocorrelation	$\text{Corr}(x_t, x_{t-1})$
Interquartile Range (IQR)	$\text{IQR} = Q_3 - Q_1$
Kurtosis	$\text{Kurtosis}(x) = \frac{1}{n}\sum_{i=1}^{n}\frac{(x_i-\mu)^4}{\sigma^4}$
Entropy	$S = -\sum_i p(x_i)\log(p(x_i))$
Energy	$\text{Energy} = \sum x_i^2$
Contrast	$\text{Contrast} = \sigma^2$
Mean Absolute Deviation	$\text{MAD} = \frac{1}{n}\sum_{i=1}^{n}\lvert x_i - \mu\rvert$
Shannon Entropy	$H = -\sum x_i \log_2(x_i)$
Range	$\max(x) - \min(x)$
Skewness	$\text{Skew}(x) = \frac{1}{n}\sum_{i=1}^{n}\frac{(x_i-\mu)^3}{\sigma^3}$
Maximum Value	$\max(x)$
Signal-to-Noise Ratio	$\text{SNR} = \frac{\mu}{\sigma}$
25th Percentile (Q_1)	$\text{Percentile}_{25} = Q_1$
50th Percentile (Q_2)	$\text{Percentile}_{50} = Q_2$ (Median)
75th Percentile (Q_3)	$\text{Percentile}_{75} = Q_3$
Harmonic Mean	$\text{HM} = \frac{n}{\sum \frac{1}{x_i}}$
Coefficient of Variation	$\text{CV} = \frac{\sigma}{\mu}$
Standard Error of Mean	$\text{SEM} = \frac{\sigma}{\sqrt{n}}$
Median Absolute Deviation	$\text{MedAD} = \text{median}(\lvert x_i - \text{median}(x)\rvert)$
Root Mean Square	$\text{RMS} = \sqrt{\frac{1}{n}\sum x_i^2}$

3.1.3. Feature selection using ZFMIS

Feature selection plays a crucial role in identifying the most relevant features while reducing noise, overfitting, and training times (Hira and Gillies, 2015). Recently, hybrid multi-stage selection strategies have also been explored, as demonstrated in Uysal (2018) for text classification. In this study, the ZFMIS a two-stage feature selection approach is developed to remove less important or irrelevant statistical features and select only informative ones.

Zero-based filtering process. The first stage of the ZFMIS feature selection approach is zero-based filtering, which eliminates features with a high proportion of zero values (sparse data) which can introduce noise into the learning process. This key step refines the statistical features by removing features with limited relevance or minimal variability. Discarding uninformative and sparse features reduces the dimensionality of the dataset while enhancing the computational efficiency. Formally, if the original set of statistical features is represented by $F_{\text{stat}} = \{x_1, x_2, \ldots, x_m\}$ then

$$F_{stat-filtered} = \left\{x \in F_{stat} | percent - zero(x) < 50\%\right\} \tag{4}$$

Whereas the $percent-zero(x)$ is used to calculate the number of zeros in n elements of each feature x, then

$$\text{percent-zero}(x) = \frac{\#\{i \mid x_i = 0\}}{n} \times 100 \tag{5}$$

where $\#\{i \mid x_i = 0\}$ denotes the number of elements in x that are equal to zero, and n is the total number of elements in x.

This means only those features are selected in which $percent - zero(x) \leq 50\%$. The 50% threshold was selected based on empirical evaluation, where different thresholds (40%, 50%, 60%, and 70%) were

tested. The 50% threshold provided the best balance between feature retention and performance improvement.

To validate the effectiveness of zero-based filtering, the features are ranked using MI scores. The removed features consistently had the lowest MI scores across all datasets, confirming their minimal contribution to the model. Additionally, a decision tree classifier is used to compare performance before and after zero-based filtering, further supporting the effectiveness of this approach. This method ensures that only features with sufficient variability and non-zero content are retained, preserving valuable information for predictive modeling.

Feature selection based on mutual information. In the second stage, the features selected after zero-based filtering undergo Mutual Importance Selection (MIS) where the features are ranked based on their MI with the target variable Y. MI measures the correlation between two variables and their interdependence (Huang et al., 2024). Let X and Y are two continuous random variables, then the $MI(X;Y)$ between them is defined as (Kraskov et al., 2004)

$$MI(X;Y) = \int_Y \int_X p_{(X,Y)}(x,y).\log(\frac{p_{(X,Y)}(x,y)}{p_X(x)p_Y(y)})dxdy \tag{6}$$

whereas $p(x,y)$ represents the joint probability distribution and $p(x)$ and $p(y)$ represent the marginal probability distribution of X and Y. So high MI means that the feature X is strongly related to Y, low MI means that the X has weak statistical dependence on the Y, whereas 0 MI means that X and Y are strictly independent.

Furthermore, a hierarchical approach is applied based on MI ranking to systematically select and evaluate the subsets of top features (e.g., top 18, 15, 12, 9, 6, 3) to find the optimal combination features for model training. The aim is to balance model complexity and interpretability while reducing computational cost and overfitting. Given a feature set ranked using MI score $F_{MIranked} = \{x_1, x_2, x_3, \dots, x_n\}$ where x_i denotes the ith feature in the ranked feature set, then according to the hierarchical selection strategy k subsets of ranked features are formed.

$$F_{select}(k) = (x_1, x_2, \dots, x_k) \tag{7}$$

Where $k \in (3, 6, 8, 12, 15, 18)$. Each subset of feature from the k subsets is separately tested by training a model on it to evaluate the contribution of these k features to the overall classification performance. The subsets are tested in descending order, starting from the subset with the highest number of top features to the subset with the lowest number of top features, to explore how removing less important features affects model accuracy and interoperability. The pseudocode for this step can be found in Algorithm 2.

Algorithm 2: Two-Stage Feature Selection Algorithm: Zero-based filtering with mutual importance selection

Input: Original feature set $F_{\text{stat}} = \{x_1, x_2, \dots, x_m\}$, target variable Y.
Output: Ranked feature set F_{ranked} based on mutual importance scores.
Step 1: Zero-Based Filtering
Initialize filtered feature set $F_{\text{stat-filtered}} \leftarrow \emptyset$.
for *each feature* x_i *in* F_{stat} **do**
  Compute $percent_zero(x_i)$, the percentage of zero values in x_i.
  if $percent_zero(x_i) \leq 50\%$ **then**
    Add x_i to $F_{\text{stat-filtered}}$.
$F_{\text{stat-filtered}} = \{x_j \in F_{\text{stat}} \mid percent_zero(x_j) \leq 50\%\}$.
Step 2: Mutual Importance Selection
for *each feature* x_i *in* $F_{stat\text{-}filtered}$ **do**
  Compute the mutual information score $MI(x_i;Y)$.
  Store $MI(x_i;Y)$.
Step 3: Ranking and Feature Selection
Rank the features $\{x_1, x_2, \dots, x_n\}$ in descending order based on $MI(x_i;Y)$.
Construct the ranked feature set $F_{\text{ranked}} = \{x_{(1)}, x_{(2)}, \dots, x_{(n)}\}$, where $x_{(i)}$ is the i-th feature in the ranking.
Step 4: Hierarchical Selection Strategy
for $k \in \{3, 6, 9, 12, 15, 18\}$ **do**
  Construct the selected feature subset:
  $F_{\text{selected}}(k) = \{x_{(1)}, x_{(2)}, \dots, x_{(k)}\}$
  Store $F_{\text{selected}}(k)$ for each value of k.

3.1.4. Classification and rule-based modeling for interpretability

This study applies two rule-based models, i.e., Decision Trees (Vasilev et al., 2020) and RuleFit (Friedman and Popescu, 2008) for the classification of final statistical features. These models generate human-understandable rules to enhance interpretability and show how features contribute to predictions. The rules are explicit "if-then" conditions that capture the relationship between features and the target classes.

Decision tree. A DT is a model that recursively splits data (statistical features) into subsets based on its values in a hierarchical manner, resulting in a tree-like structure. The nodes represent the conditions used for feature classification, whereas branches represent the outcomes of those conditions. The final nodes, also called leaf nodes, represent the predicted classes. By default, DT uses the Gini impurity measure (Tangirala, 2020) for feature splitting at the nodes. DTs directly map the decision path from the root node to the leaf nodes (specific prediction), forming interpretable rules.

RuleFit. RuleFit is a rule-based ensemble model that combines the strengths of rule-based learning and linear modeling, capturing both linear and complex interactions within data. It uses DTs and linear regression to identify non-linearities and linear links. The DTs generate decision rules, and the regression model optimizes performance and prevents overfitting. RuleFit balances expressive power and interpretability, making it ideal for applications requiring clear model insights. The model generates a set of rules from the dataset's features, evaluates each rule's performance, selects the most relevant rules, and then applies the most appropriate rules to predict or classify new samples based on their features. RuleFit generates rules with the default gradient boosted trees ensemble model. The Least Absolute Shrinkage and Selection Operator (LASSO) linear regressor is used to identify the most important rules for predictions by assigning weights to the extracted rules.

3.2. Hyper-parameters optimization for custom MobileNetV2, Decision tree, and Rulefit

This study uses the Grid Search technique for hyperparameter optimization to evaluate various combinations of hyperparameters systematically. The Grid search selects optimal hyperparameter values based on the highest validation accuracy achieved during model training, enhancing the performance of the custom MobilNetV2 model, DT, and RuleFit models. This study considers key hyperparameters crucial for classification, such as learning rate, number of epochs, tree size, depth, etc. However, due to limited computational resources, a small subset of image datasets is used to find optimal hyperparameters for the custom MobileNetV2 model. In contrast, for ML models, i.e., DT and RuleFit statistical features are used. Table 2 summarizes the ranges of hyperparameters explored for each model in this study.

3.3. Visual explanation using statistical feature map overlay visualization

This study uses a novel Grad-CAM-inspired statistical feature map overlay visualization (SFMOV) technique to generate heat maps and enhance transparency. The SFMOV pseudocode is presented in Algorithm 3.

SFMOV does not use activations like traditional approaches, including Grad-CAM, but uses statistical measures, i.e., mean, skewness,

Table 2
Hyperparameters values for each model used in grid search optimization.

Model	Hyperparameter	Values
Custom MobileNetV2	Number of Epochs	10, 30, 50
	Batch Size	16, 32, 64, 128
	Learning Rate	0.001, 0.005, 0.01, 0.02
Decision tree	Criterion	Entropy, Gini
	Max Depth	5, 10, 20, 30
	min_samples_leaf	1, 2, 4
	min_samples_split	2, 5, 10
RuleFit	Tree Size	3, 5, 7
	Sampling Fraction	0.7, 0.8, 1.0
	Max Rules	25, 100, 200, 500, 1000, 2000, 3000

entropy, and a combination of these measures to visualize the most important regions for classification within medical images. The statistical feature maps are computed per pixel (x, y) for each color channel. Specifically, the mean map $M_c(x, y)$ represents the mean pixel intensity at location (x, y) across all channels (c), providing a brightness-based interpretation of the image. The skewness map $S_c(x, y)$ captures asymmetry in pixel distributions, while the entropy map $E_c(x, y)$ emphasizes texture complexity.

To provide a more comprehensive visualization, a combined heatmap $H(x, y)$ is constructed as:

$$H(x, y) = w_1 M_c(x, y) + w_2 S_c(x, y) + w_3 E_c(x, y) \tag{8}$$

where w_1, w_2, and w_3 are the dense values corresponding to mean, skewness, and entropy, respectively. This integrated heatmap leverages multiple statistical indicators to highlight diagnostically relevant regions more effectively than any individual map alone. While the mean heatmap primarily reflects overall intensity, the combined heatmap incorporates asymmetry and texture complexity, ensuring that subtle but clinically significant structural variations are captured.

Eq. (8) provides a structured integration of multiple statistical measures, allowing for a more robust and informative visualization.

The selection of mean, skewness, and entropy as the key statistical features in the proposed SFMOV method is because they have the ability to represent the inherent characteristics of medical image regions in a comprehensible and effective manner. Specifically, the mean characterizes the overall brightness or intensity distribution and indicates the overall tissue density; skewness is a quantitative measure of asymmetry in pixel intensity distributions and can be employed to reveal abnormal or aberrant structures; and entropy defines texture complexity and indicates regions with heterogeneous patterns typically associated with pathological alteration.

This combination provides a complementary and integral knowledge of global and local image characteristics, allowing for finer visualization of diagnostically relevant regions. By integrating these three descriptors, the method enhances interpretability, allowing clear differentiation between critical regions without relying on complex statistical metrics. On the other hand, other measures such as harmonic mean or autocorrelation, as logically valid as they are, often fail to allow intuitive visual interpretation, thus becoming less accessible to clinicians or researchers to directly correlate with anatomical or pathological significance. Therefore, focusing on mean, skewness, and entropy ensures that the resulting visual explanations are not only meaningful but also accessible and practically useful for understanding model decisions.

This approach has two stages as shown in Fig. 2. In the first stage, it calculates the statistical properties (mean, skewness, and entropy) from each channel in the convolutional layer of the custom MobileNetV2 model. In the second stage, the dense layer statistical properties (mean, skewness, and entropy) are used as weights for the convolutional maps. Finally, this method overlays these statistical measures onto the original image using a jet color scheme, where red represents the most critical and highly activated regions, and blue signifies less significant or inactive areas. The jet color scheme effectively highlights the crucial regions for enhanced interpretability.

Algorithm 3: Statistical Feature Map Overlay Visualization (SFMOV)

Input: Input Image $I \in \mathbb{R}^{H\times W\times C}$, trained custom MobileNetV2 Model $\mathcal{CMM}$.
Output: Heatmaps for Mean, Skewness, Entropy, and Combined Weighted Overlay.

Step 1: Load and Preprocess Image

- **Step a: Image resizing**
  $I_{\text{res}} \in \mathbb{R}^{H\times W\times 3}$
  The original image I_{orig} is resized (I_{res}) to the target dimensions $H \times W$.
- **Step b: Normalize the Image (scale the image pixel values to the range** $[0, 1]$**)**
  $\mathbf{I}(x, y, c) = \frac{I_{\text{res}}(x,y,c)}{255}, \quad \forall x \in \{1, \ldots, H\}, y \in \{1, \ldots, W\}, c \in \{1, 2, 3\}$
  $\mathbf{I} \in [0, 1]^{H\times W\times 3}$

Step 2: Extracting convolutional feature maps and dense layer activations of $\mathcal{CMM}$

- **Step a: Calculate dense layer statistics**
  $w1 = dense_mean$
  $w2 = dense_skewness$
  $w3 = dense_entropy$
- **Step b: Local Convolutional Feature Maps Statistics:**
  for *each color channel* $c \in \{R, G, B\}$ **do**
  Compute the following statistical maps for channel c:
  - **Mean Map** $M_c(x, y) = \text{mean}(I_c(x, y))$
  - **Skewness Map:** $S_c(x, y) = \text{skewness}(I_c(x, y))$
  - **Entropy Map:** $E_c(x, y) = \text{entropy}(I_c(x, y))$

  Step c: Normalization and Resizing: Normalize and resize each map: $\text{map}_{\text{norm}} = \frac{\text{map}-\min(\text{map})}{\max(\text{map})-\min(\text{map})+\epsilon}$

Step 3: Combine Maps
Compute a weighted combination of $M_c(x, y)$, $S_c(x, y)$, and $E_c(x, y)$ to create a composite heatmap:

$$H(x, y) = w_1 M_c(x, y) + w_2 S_c(x, y) + w_3 E_c(x, y)$$

Step 4: Overlay Visualization
For visualization, overlay the combined heatmap $H(x, y)$ onto the original image I.

4. Results and discussion

This section introduces information about the experiments performed and the results produced in this study. This section also provides additional information about the datasets, experimental setup, optimized hyper-parameters, and evaluation metrics used in this study.

4.1. Datasets

This study uses five publicly available Kaggle datasets with various medical imaging modalities to assess the proposed approach's generalizability and robustness across distinct medical conditions. The five datasets are well-known datasets, and each covers diverse diseases, imaging techniques, and anatomical regions. The summary of the datasets is given in Table 3 whereas the representative samples from these datasets are given in Fig. 3.

4.1.1. COVID-19 Radiography Database (Dataset 1)

The first dataset used in this study is the "COVID-19 Radiography Database" (Rahman et al., 2022), a multiclass dataset consisting of 21,165 chest radiograph images. The dataset is created by a team of researchers from different countries including Pakistan, Qatar, Bangladesh, and Malaysia. The dataset contains images of four classes with their corresponding masks. The recent version of the dataset

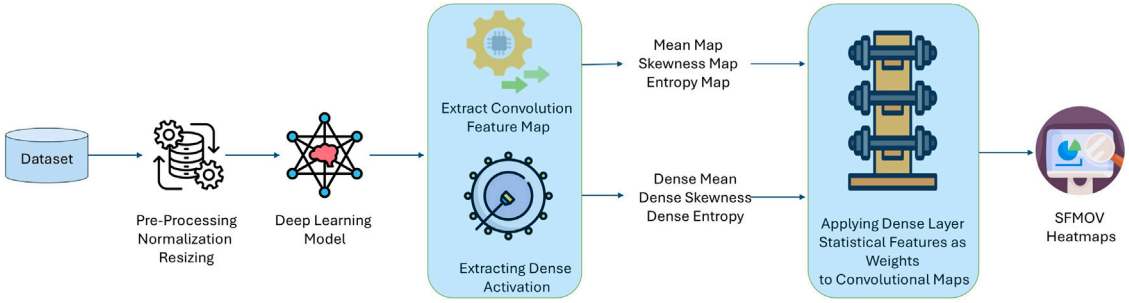


Fig. 2. SFMOV process, extracting Mean, Skewness, and Entropy maps from convolutional features and dense layer activations, normalizing and combining them into a weighted heatmap overlaid on the input image.

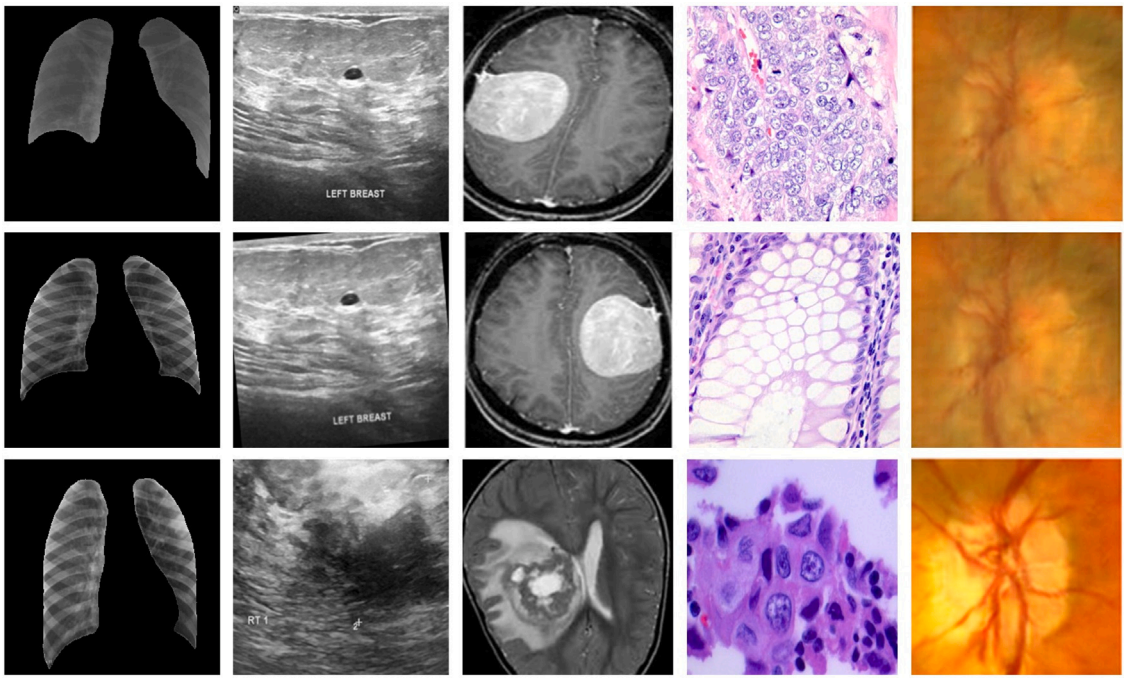


Fig. 3. Representative samples from all datasets, first, second, third, fourth, and fifth columns have representative samples from dataset 1, dataset 2, dataset 3, dataset 4, and dataset 5, respectively.

consists of 10,192 normal, 3616 COVID-19, 1345 viral pneumonia, and 6012 lung opacity chest radiograph images.

4.1.2. Ultrasound Breast Images for Breast Cancer (Dataset 2)

The second dataset used in this study is "Ultrasound Breast Images for Breast Cancer" (Sairam, 2024). This dataset used rotation and sharpening augmentation techniques and contains 9016 ultrasound images after augmentations with 4574 benign and 4442 malignant breast cancer images.

4.1.3. BTTypes (Dataset 3)

The third dataset used in this study is "BTTypes" (Siddiqui, 2024). The images of this dataset are taken from another publicly available dataset, "Brain MRI Images for Brain Tumor Detection", (Sabale, 2024) which has 11 benign MRI images and 12 malignant MRI images. The authors of BTTypes applied augmentations to the original dataset and created a total of 2400 MRI images after augmentation, with 1200 benign MRI images and 1200 malignant MRI images.

4.1.4. Lung and Colon Cancer Histopathological Images (Dataset 4)

The fourth dataset is the "Lung and Colon Cancer Histopathological Images" dataset (Borkowski et al., 2019), which contains 25,000 images in five classes: lung benign tissue, lung adenocarcinoma, lung squamous cell carcinoma, colon adenocarcinoma, and colon benign tissue and each class contains 5000 images.

4.1.5. ACRIMA from Generated Eye Dataset for Glaucoma Detection (Dataset 5)

The fifth dataset used in this study is from the Kaggle repository "Generated Eye Dataset for Glaucoma Detection" (Saud, 2024) which contains five datasets (RIM-ONE, ACRIMA, HRF, Drishti-GS, and ORIGA-LIGHT) for glaucoma detection and 30,000 retinal fundus images in total. ACRIMA dataset (Diaz-Pinto et al., 2019) is used in this study, which contains 6000 retinal fundus images with 3000 glaucoma and 3000 normal images.

4.2. Experimental setup

This study uses a rigorous experimental setup to verify the effectiveness and systematic evaluation of the proposed approach across multiple medical imaging datasets. The components of the experimental setup i.e., hardware configuration, software environment, and evaluation metrics are carefully chosen to address the computational requirements of training different models such as custom MobileNetV2, DT, and RuleFit models. Table 4 gives detailed information about the experimental setup.

Table 3
Summary of the datasets used in this study.

Dataset	# of classes	# of images	Class names	Imaging modality	File format	Resolution
COVID-19 Radiography Database	4	21,165	Normal, Lung Opacity, Viral Pneumonia, COVID	X-ray	PNG	299 × 299
Ultrasound Breast Images for Breast Cancer	2	9016	Benign, Malignant	Ultrasound	PNG	227 × 227, 224 × 224
BTTypes	2	2400	Benign, Malignant	MRI	PNG	224 × 224
Lung and Colon Cancer Histo-pathological Images	5	25 000	Lung benign tissue (lung_n), Lung adenocarcinoma (lung_aca), Lung squamous cell carcinoma (lung_scc), Colon adenocarcinoma (colon_aca), Colon benign tissue (colon_n)	Histo-pathology	JPEG	768 × 768
ACRIMA from Generated Eye Dataset for Glaucoma Detection	2	6000	Glaucoma, Normal	Retinal Fundus	PNG	128 × 128

Table 4
Experimental setup: Software and hardware specifications.

Category	Specification
Hardware Configuration	
Processor	Intel(R) Core(TM) i7-8650U CPU @ 1.90 GHz 2.11 GHz (8 Cores)
RAM	16 GB
Storage	500 GB SSD
Software Environment	
Programming Language	Python 3.12
Operating System	Windows 11 Pro
Data Processing Libraries	Pandas 2.2, NumPy 1.26, OpenCV 4.10
DL Framework	Keras 3.5, TensorFlow 2.17
Visualization Tools	Seaborn 0.13, Matplotlib 3.9
Evaluation Metrics	
Model Selection Metrics	Training Loss, Validation Loss
Classification Metrics	Accuracy, Precision, Recall, F1-Score
Explainability	Visualization of important regions
Interpretability	DT rules, RuleFit rules

4.3. Optimized hyperparameters for each dataset

To enhance the performance of MobileNet, DT, and RuleFit models the hyperparameter optimization is performed separately for each dataset to address the differences in data structure across datasets. The hyperparameter values that maximize the model's effectiveness on each dataset are identified using the grid search technique. Table 5 summarizes the optimized hyperparameters for MobileNet, Decision Tree, and RuleFit models on each dataset.

4.4. Classification results

This section provides a thorough examination of classification performance across five different medical imaging datasets—each representing a different diagnostic task. Table 5 shows the optimized hyperparameters used for each dataset. The approach starts by converting deep features extracted using a custom MobileNetv2 model into statistical features. Then, two-stage feature selection, i.e., ZFMIS, is applied to select the most important feature hierarchically. The effect of feature selection on DT and RuleFit performance is then assessed using a comprehensive statistical feature set.

The comprehensive statistical features set (26 features) is an extensive set that encompasses a broad range of statistical features, including percentile 25, signal-to-noise, autocorrelation, kurtosis, Shannon entropy, and other derived statistical features (Table 1). A two-stage feature selection approach is applied here; zero-based filtering was applied to remove features with more than 50% zero values, resulting in feature subsets with 26, filtered, 18, 15, 12, 9, 6, and 3 features. After defining these feature sets, the classification performance is evaluated using DT and RuleFit models across varying feature counts. To provide insights into the key features for rule extraction and interpretability, the top three features for each dataset based on their MI values are given in Table 6.

On all datasets except Dataset 5 (ACRIMA from Generated Eye Dataset for Glaucoma Detection), after the first stage of feature selection, i.e., zero-based filtering, median, median_abs_dev, min_value, percentile_25, and percentile_50 are removed from the 26-feature set.

Whereas, on "ACRIMA from Generated Eye Dataset for Glaucoma Detection dataset" two statistical features i.e., min_value and percentile_25 are removed from the 26 feature set. The findings of each dataset are shown in the sections that follow, along with tables that show the classification performance achieved by each model for every feature subset. To shed light on the ideal configurations, the most significant results from every dataset are highlighted.

Furthermore, the variation of Decision Tree (DT) and RuleFit performance on various datasets results from their internal learning processes. DT is a tree model and is extremely sensitive to the individual structure

Table 5
Summary of the optimized hyperparameters on each dataset.

Dataset	Custom MobileNetV2	Decision tree	Rulefit
COVID-19 Radiography Database	learning_rate: 0.001 epochs: 50 batch_size: 128	criterion: entropy max_depth: 10 min_samples_leaf: 4 min_samples_split: 2	tree_size: 3 sample_fraction: 0.7 max_rules: 3000
Ultrasound Breast Images for Breast Cancer	learning_rate: 0.001 epochs: 10 batch_size: 16	criterion: entropy max_depth: 10 min_samples_leaf: 4 min_samples_split: 2	tree_size: 3 sample_fract: 0.7 max_rules: 2000
BTTypes	learning_rate: 0.001 epochs: 10 batch_size: 16	criterion: entropy max_depth: 5 min_samples_leaf: 4 min_samples_split: 2	tree_size: 3 sample_fract: 0.7 max_rules: 2000
Lung and Colon Cancer Histopathological Images	learning_rate: 0.001 epochs: 50 batch_size=32	criterion: entropy max_depth: 10 min_samples_leaf: 4 min_samples_split: 2	tree_size: 3 sample_fract: 0.7 max_rules: 2000
ACRIMA from Generated Eye Dataset for Glaucoma Detection	learning_rate: 0.001 epochs: 30 batch_size: 64	criterion: gini max_depth: 10 min_samples_leaf: 1 min_samples_split: 10	tree_size: 5 sample_fract: 0.7 max_rules: 100

Table 6
Top three selected features from 26 features set for each dataset based on MI after employing two-stage feature selection algorithm (ZFMIS).

Dataset	Top feature 1 (MI score)	Top feature 2 (MI score)	Top feature 3 (MI score)
COVID-19 Radiography Database	HM (0.61)	Skew (0.60)	CV (0.57)
Ultrasound Breast Images for Breast Cancer	mean (0.23)	H (0.22)	GM (0.22)
BTTypes	HM (0.42)	GM (0.41)	entropy (0.35)
Lung and Colon Cancer Histopathological Images	corr (0.85)	SNR (0.61)	CV (0.61)
ACRIMA from Generated Eye Dataset for Glaucoma Detection	max (0.30)	range (0.30)	σ (0.30)

of a dataset. When the dataset consists of clear hierarchical structures and obvious feature splits, DT operates effectively. However, DT has an overfitting weakness, especially when handling noisy or interactive relationships of features. In contrast, RuleFit produces multiple decision rules and combines them in a sparse linear model. This allows RuleFit to generalize better on data sets with non-linearly separable decision boundaries by one tree structure. In some cases, though, a linear combination of rules in RuleFit may decrease its capacity to capture complex interactions among features across a deep decision tree. These differences explain the variation in DT and RuleFit performance observed between the datasets used in this study.

4.4.1. Classification results on COVID-19 Radiography Database

The results of the COVID-19 Radiography Database are reported and discussed in this section. From the analysis of Table 7, it can be concluded that DT and RuleFit models achieved high classification performance on this dataset, with no significant decrease in classification performance as the feature count decreased to optimized levels. The DT classifier achieved the highest classification performance with accuracy, precision, recall, and F1-score of 88.93%, 89.18%, 88.90%, and 88.88%, respectively on a comprehensive 26-feature set. Similarly, RuleFit demonstrated a remarkable performance with the highest accuracy of 87.26% using 26 features from a 26-feature set.

In summary, the experiments' classification findings demonstrate that although a larger feature set (26 features) might enhance model performance, carefully chosen, smaller sets can still produce satisfactory classification performance, and different feature configurations are advantageous for different classifier types. This harmony between feature selection and model accuracy supports the feasibility of both classifiers for categorizing radiographic data.

4.4.2. Classification results on Ultrasound Breast Images for Breast Cancer

The Ultrasound Breast Images for Breast Cancer results are summarized in Table 8. Thus, this table shows results using the 26-feature set. These results demonstrate how varying feature sets impact the classification performance of RuleFit and DT. Both DT and RuleFit classifiers exhibit a decreasing trend of classification performance across all metrics (accuracy, precision, recall, and F1-score) on the 26 feature set (Table 8) except DT on the filtered (21) features and 6 features, the DT which achieved its best accuracy of 81.53% with a filtered set of 21 features compared to the accuracy of 80.80% with all 26 features and the highest accuracy of 77.53% with 3 features compared to the accuracy of 77.04% with 6 features. The results clearly show that the RuleFit model consistently outperforms the DT model across all feature sets, achieving its highest accuracy of 85.29% using the 26 features. Comparatively, the DT model achieves its best accuracy of 81.53% with a filtered set of 21 features.

Compared to DT, RuleFit appears less sensitive to feature reduction, demonstrating a greater capability to retain classification performance with a larger number of features. From these results, we conclude that RuleFit achieves optimal accuracy and consistency in predictive performance and is more suitable for handling complex feature interactions within this dataset.

4.4.3. Classification results on Brain MRI Images for Brain Tumor Detection

This section analyzes the results of the Brain MRI Images for Brain Tumor Detection dataset. The classification performance from Table 9 reveals insightful patterns when comparing RuleFit and DT classifiers with varying numbers of statistical features, considering the comprehensive features set.

Table 7
Classification results of COVID-19 Radiography Database on a comprehensive set of statistical features with varying feature counts for DT and RuleFit.

Classifier	Selected features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Decision Tree	26	88.33	88.72	88.29	88.28
	Filtered (21)	**88.93**	**89.18**	**88.90**	**88.88**
	18	88.88	89.13	88.85	88.83
	15	88.45	88.74	88.44	88.40
	12	88.22	88.51	88.21	88.17
	9	83.81	84.58	83.90	83.56
	6	74.44	76.40	74.54	73.95
	3	67.20	69.74	67.37	66.29
RuleFit	26	87.26	**88.02**	87.20	87.24
	Filtered (21)	87.10	87.79	87.04	87.07
	18	86.98	87.64	86.93	86.95
	15	**87.21**	87.87	**87.16**	**87.18**
	12	86.59	87.54	86.51	86.55
	9	82.29	83.23	82.21	82.21
	6	78.71	79.94	78.63	78.58
	3	69.38	69.60	69.33	69.06

Table 8
Classification results of Ultrasound Breast Images for Breast Cancer on a comprehensive set of statistical features with varying feature counts for DT and RuleFit.

Classifier	Selected features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Decision Tree	26	80.80	80.82	81.32	80.73
	Filtered (21)	**81.53**	**81.55**	82.05	**81.46**
	18	80.72	80.57	81.04	80.60
	15	78.34	78.44	78.89	78.28
	12	75.65	76.58	76.72	75.65
	9	75.49	76.50	76.60	75.49
	6	77.04	77.13	77.56	76.97
	3	77.53	70.97	**82.24**	76.19
RuleFit	26	**85.29**	**85.16**	**85.71**	**85.21**
	Filtered (21)	84.97	84.83	85.37	84.88
	18	83.25	83.14	83.66	83.16
	15	82.52	82.51	83.03	82.45
	12	79.98	79.96	80.45	79.90
	9	79.25	79.37	79.83	79.19
	6	80.64	80.62	81.11	80.56
	3	78.35	78.39	78.85	78.27

Using a comprehensive set of 26 statistical features, RuleFit achieved the best accuracy of 92.50% with 21 features, i.e., filtered features (features after removing the features with more than 50% of 0s). The classification performance almost remains stable with further feature selection i.e., with 18, 15, 12, and 9 features, but declined below 6 features. On the same feature set, the DT classification performance is improved with feature selection achieving 90.83% accuracy with 21, 12, and 9 features. However, accuracy dropped with 6 or fewer features, indicating DT's need for a larger set to maintain accuracy. Overall, Both models perform well with 9–12 features, balancing accuracy and simplicity. RuleFit consistently outperformed DT, especially with fewer features, reflecting its advantage in interpretability and performance.

4.4.4. Classification results on Lung and Colon Cancer Histopathological Images

The results obtained for the Lung and Colon Cancer Histopathological Images dataset are shown in Table 10. Considering the comprehensive features set, this table reveals insightful patterns when comparing RuleFit and DT classifiers with varying numbers of statistical features.

The results of Table 10 demonstrate that the DT tree has achieved the highest accuracy of 94.10% using 18 top features from the comprehensive 26 features set, indicating that a comprehensive set of statistical features enables a robust classification model. Whereas, RuleFit achieves the highest accuracy of 87.36% using all 26 features of the comprehensive features set.

Furthermore, it is worth noting that on a comprehensive feature set reducing the features to a filtered subset of 21 and down to 12 and 9 does not significantly affect the performance of DT, with accuracy values staying above 92%.

Overall, the results of DT and RuleFit (Table 10) suggest that both classifiers achieve optimal performance on a larger feature set. The DT achieves higher classification performance consistently outperforming the RuleFit across different feature sets. These results indicate that the DT model is better suited for the statistical features of the dataset. The DT classification results indicate that the DT can capture more distinct patterns with a larger feature set resulting in stronger model performance. Furthermore, results after hierarchical feature selection demonstrate that less informative features can be removed without a significant drop in performance. These results highlight the significance of establishing a balance between information reduction and classification performance of the model, especially when dealing with interpretable models such as RuleFit and DT that perform best when given a reasonably extensive feature set.

4.4.5. Classification results on ACRIMA from Generated Eye Dataset for Glaucoma Detection

The classification results of the ACRIMA from the Generated Eye Dataset for Glaucoma Detection dataset are presented in Table 11. Using a comprehensive set of features the classification performance of DT is further improved to 81.44% (Table 11) with 18 selected features highlighting that an optimal subset of features can improve model performance. Interestingly, the performance decreases using features more than 18 or less than 18 because of overfitting or the inclusion of less relevant features even though using a feature set of 24 does not impact performance and remains at 80.78%, suggesting the removal of redundant or irrelevant features.

Table 9
Classification results of Brain MRI Images for Brain Tumor Detection on comprehensive set of statistical features with varying feature counts for DT and RuleFit.

Classifier	Selected features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Decision Tree	26	90.00	90.00	90.18	89.99
	Filtered (21)	**90.83**	**91.19**	**91.22**	**90.83**
	18	90.55	90.65	90.81	90.55
	15	90.56	90.65	90.81	90.55
	12	**90.83**	90.82	91.00	90.82
	9	**90.83**	90.82	91.00	90.82
	6	88.33	88.94	88.80	88.33
	3	89.16	89.12	89.29	89.15
RuleFit	26	91.67	91.63	91.82	91.65
	Filtered (21)	**92.50**	**92.52**	**92.71**	**92.49**
	18	91.67	91.71	91.89	91.66
	15	91.67	91.67	91.85	91.66
	12	92.22	92.22	92.41	92.21
	9	92.22	92.22	92.41	92.21
	6	89.17	89.43	89.51	89.17
	3	89.17	89.36	89.47	89.16

Table 10
Classification results of Lung and Colon Cancer Histopathological Images on the comprehensive set of statistical features with varying feature counts for DT and RuleFit.

Classifier	Selected features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Decision Tree	26	94.03	94.12	93.96	93.98
	Filtered (21)	94.05	94.15	93.98	94.00
	18	**94.10**	**94.20**	**94.03**	**94.05**
	15	93.87	93.87	93.80	93.80
	12	94.05	94.12	93.98	93.98
	9	92.64	92.70	92.56	92.57
	6	87.65	87.97	87.49	87.37
	3	80.42	80.74	80.20	79.84
RuleFit	26	**87.36**	**88.35**	**87.35**	**87.27**
	Filtered (21)	87.04	88.13	87.03	86.92
	18	87.04	88.11	87.03	86.93
	15	86.61	87.80	86.61	86.48
	12	86.61	87.76	86.60	86.50
	9	86.16	87.19	86.15	86.08
	6	80.40	81.69	80.34	80.40
	3	81.23	82.04	81.17	81.09

Similarly, RuleFit achieved the same accuracy of 83.78% on both the 26 feature set and the filtered feature set (24 features) means that removing less important features does not degrade performance. This suggests that RuleFit is effective with the selected optimal subset of features or a larger feature set.

Overall, the RuleFit demonstrates superior classification performance as compared to DT across all feature sets. This highlights the power of ensemble methods and their strength in leveraging both classification performance and interpretability. Interestingly, both classifiers achieved the best classification performance with intermediate selected features (9 to 18). The stable performance of both models with filtered feature sets demonstrates that removing irrelevant features does not compromise accuracy. Also, the results proved that larger feature sets with additional information, and redundant or irrelevant features can negatively impact performance. Furthermore, these findings emphasize the role of feature engineering and feature selection (identification of the optimal subset of features) in classification tasks.

4.5. Rule extraction for interpretability

This study further explored rule extraction using DT and RuleFit after achieving satisfactory classification results on a diverse medical image dataset using these models to enhance interpretability and transparency. This transparency is necessary, especially for medical imaging data to ensure that the decisions made by these classifiers are understandable and can be validated by users or domain experts. Rule extraction in this study aims to extract simple, human-understandable, and interpretable conditions that explain the relationship between specific feature values and classification outcomes.

In this study, a comprehensive set of 26 statistical features is used for all datasets, so using and evaluating them all at once in rules might result in complicated rules that could be harder to comprehend. On the other hand, using only the top three selected features after the hierarchical feature selection approach provides more understandable rules while yet maintaining satisfactory classification performance. Therefore, to balance classification performance and interpretability, this work focuses on rule extraction utilizing three statistical features.

Furthermore, the rules generated from the DT and RuleFit models are simplified to reduce redundancy and improve interpretability. The strictest bounds are retained for each feature, ensuring minimal overlap between conditions. For example, if a feature has conditions like "$GM > 2$" and "$GM > 3$", the less restrictive condition is discarded, and only "$GM > 3$" is retained. This approach eliminates logical redundancies and maintains only the most restrictive thresholds for each feature. The rules, presented in a simplified "If-Then" format and prioritized by significance scores, make it easier to identify the model's most important decision-making processes. The simplification process is widely recognized in the field of rule-based machine learning and follows standard practices for enhancing rule interpretability.

In this study, to enhance simplicity and clarity, rules for two representative datasets on which more than 90% classification accuracy is achieved are presented instead of all five datasets. The selected datasets (Brain MRI Images for Brain Tumor Detection and Lung and Colon Cancer Histopathological Images) are representative of the model's ability to generalize across diverse medical imaging data while maintaining high classification accuracy. Furthermore, thousands of potential rules are produced by both the DT and RuleFit models after the execution of

Table 11
Classification results of ACRIMA from Generated Eye Dataset for Glaucoma Detection on the comprehensive set of statistical features with varying feature counts for DT and RuleFit.

Classifier	Selected features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Decision Tree	26	80.78	80.71	80.76	80.73
	Filtered (24)	80.78	80.71	80.77	80.73
	18	**81.44**	**81.39**	**81.48**	**81.41**
	15	74.11	74.51	74.43	74.11
	12	74.55	74.77	74.79	74.56
	9	72.33	72.99	72.75	72.31
	6	72.89	73.43	73.26	72.88
	3	72.77	73.33	73.16	72.76
RuleFit	26	83.78	83.75	83.87	83.76
	Filtered (24)	83.78	83.75	83.87	83.76
	18	**84.56**	**84.53**	**84.64**	**84.54**
	15	76.44	76.91	76.79	76.44
	12	75.22	75.88	75.64	75.20
	9	7533	76.01	75.75	75.31
	6	74.67	75.64	75.17	74.62
	3	74.67	75.64	75.17	74.62

our approach on each medical imaging dataset. Therefore, to ensure relevance and clarity in the results, careful consideration is applied to select the most interpretable and impactful rules for presentation. Therefore, rules from both DT and RuleFit are selected based on their importance scores, which quantify the contribution of each rule to the model's predictions. Rules with greater importance scores demonstrate stronger and more persistent connections between features and the target class, making them valuable for understanding model behavior and identifying decision-making patterns. The importance score is calculated as the product of the absolute value of the rule's coefficient and support values. The importance of each rule r is calculated as:

$$\text{importance}_r = |\text{coef}_r| \times \text{support}_r \tag{9}$$

where coef_r is the absolute value of the rule's coefficient, indicating its predictive power, and support_r is the fraction of samples where the rule holds true, reflecting its relevance. Higher importance scores indicate rules with greater predictive strength and applicability in the dataset. To provide a balanced representation, the rules for each class are selected based on their importance scores. The two most significant rules are chosen for each class. This method provides interpretable information on the key feature patterns affecting every class's predictions.

Furthermore, it is important to note that the obtained rules are based on statistical measures such as corr, CV, and SNR. These features capture patterns in image texture and intensity that could be related to tumor characteristics. However, these features are not directly interpretable from a clinical perspective, they provide quantitative measures that may help distinguish between tumor types. As an example, SNR is a common measure of image quality, with higher values representing potentially higher quality images with lower noise, which could be used to discriminate between tumor features and background noise. Correlation as a texture feature detects patterns in the relationship between pixel intensity and can be reflective of tissue homogeneity. Variations in correlation could be a reflection of differences in tissue structure that may distinguish benign from malignant tumors, but such correlations must be confirmed by clinical experience.

The entropy, skewness, and mean enhance the ability to highlight important areas and have distinct visual effects on image regions. Entropy distinctly brings out complex textures, skewness shows asymmetric structures, and mean illustrates the general brightness or density in an area. Using a combination of these three features provides a comprehensive and more differentiated visualization. Furthermore, more complex features such as harmonic mean or autocorrelation, are harder to interpret visually, as their impact on an image may not clearly distinguish crucial areas. Hence, visualizing the most impactful features will highlight the key regions truly influential for the model.

4.5.1. Rules from DT

Decision trees produce hierarchical, straightforward, and interpretable rules based on split data using feature thresholds. The initial rules obtained from DT are complex and less readable, containing redundant conditions, and the same features are repeated multiple times with slightly varying thresholds. Therefore, the raw rules extracted from the trained DT model are simplified to enhance the interpretability of DT rules and reduce the complexity of rules without compromising the model's predictive accuracy. Table 12 represents the top two rules based on importance extracted using DT for each class for the two top datasets based on the classification performance.

4.5.2. Rules from RuleFit

RuleFit generates rules using a mixture of linear regression and tree-based methods. It selects the most important rules from a large number of candidate rules produced using DTs based on regularization. These rules are more compact and effective and incorporate linear terms for improved performance. These rules add depth by incorporating combinations of features, capturing more detailed relationships that may not be explicitly represented in DT splits alone. Similar to DT, the rules from the RuleFit are also simplified to generate more concise and interpretable rules. This simplification preserves prediction accuracy while ensuring that the rules are clear and concise. This approach improves the rules' interpretability and usefulness, making them suitable for both practical and research uses. Table 13 represents the two top rules extracted using DT for each class for the two top datasets based on the classification performance.

However, the relatively low importance scores in RuleFit (Table 13) are the result of its operational mechanism. RuleFit generates a compact set of rules from weak decision trees and assigns coefficients to them using a sparse linear model with L1 regularization. Therefore, the rules have small coefficients, and when multiplied by their respective support values, the total importance scores are reduced. In addition, RuleFit rules are generally more specific and cover fewer samples, which reduces their support and, therefore, their importance scores. Despite the lower importance measures, the rules derived are still applicable to feature interactions and model decisions since RuleFit can capture fine-grained patterns through an additive effect of multiple low-impact rules.

To demonstrate the strengths of both the DT and RuleFit model rules are presented for both in this study. The rules extracted from both models highlight different aspects of the data and comprehensively explain the patterns found in a simplified and human-readable way. The rules using only three statistical features improve interpretability and reduce complexity while maintaining classification performance. This approach helps bridge the gap between complex black-box models and interpretable ML, allowing for a clearer understanding

Table 12
Decision tree rules for top two datasets based on classification performance.

Dataset	Rule	Predicted class	Importance score
Dataset 3	$S \leq 3.94$	Benign	2.91
	$GM \leq 1.00 \times 10^{-3}$ & $S \leq 3.93$	Benign	2.87
	$S > 3.95$ & $GM \leq 1.00 \times 10^{-3}$	Malignant	1.99
	$GM > 1.00 \times 10^{-3}$	Malignant	3.76
Dataset 4	$corr > -8.40 \times 10^{-2}$ & $corr \leq -8.20 \times 10^{-2}$ & $SNR > 0.76$ & $CV > 1.32$	Colon adenocarcinoma	4.89
	$corr > -5.30 \times 10^{-2}$ & $corr \leq -4.40 \times 10^{-2}$ & $CV > 1.34$ & $CV \leq 1.37$ & $SNR \leq 0.76$	Colon adenocarcinoma	4.83
	$corr > -1.70 \times 10^{-2}$ & $corr \leq -1.60 \times 10^{-2}$ & $CV > 1.37$ & $SNR > 0.72$	Colon benign	4.89
	$corr > -0.11$ & $corr \leq -0.10$ & $STN > 0.75$ & $CV > 1.32$	Colon benign	4.83
	$corr > 6.0 \times 10^{-3}$ & $corr \leq 2.6 \times 10^{-2}$ & $CV > 1.29$ & $CV \leq 1.37$ & $STN > 0.78$ & $STN \leq 0.82$	Lung adenocarcinoma	3.45
	$corr > -0.08$ & $corr \leq -0.07$ & $CV > 1.44$ & $STN \leq 0.66$	Lung adenocarcinoma	3.00
	$corr > -5.20 \times 10^{-2}$ & $corr \leq -4.80 \times 10^{-2}$ & $CV > 1.37$ & $SNR > 0.67$ & $SNR \leq 0.69$	Lung benign	4.89
	$corr > -0.08$ & $corr \leq -0.07$ & $CV > 1.47$ & $STN > 0.67$ & $STN \leq 0.68$	Lung benign	3.61
	$corr > -1.42 \times 10^{-1}$ & $corr \leq -1.39 \times 10^{-1}$ & $SNR > 0.78$ & $CV \leq 1.31$	Lung squamous cell carcinoma	4.89
	$corr > -0.23$ & $corr \leq -0.17$ & $STN > 0.79$ & $STN \leq 0.81$ & $CV > 1.19$	Lung squamous cell carcinoma	3.51

Table 13
RuleFit rules for top two datasets based on classification performance.

Dataset	Rule	Predicted class	Importance score
Dataset 3	$S \leq 3.94$	Benign	0.05
	$GM > 1.00 \times 10^{-3}$ & $S > 3.88$ & $S \leq 3.92$	Benign	0.03
	$GM \leq 1.00 \times 10^{-3}$	Malignant	0.04
	$GM > 1.00 \times 10^{-3}$ & $S \leq 3.88$	Malignant	0.02
Dataset 4	$corr > -0.06$ & $CV > 1.38$	Colon adenocarcinoma	0.17
	$corr \leq 1.41$ & $corr \leq -0.17$	Colon adenocarcinoma	0.17
	$STN > 0.74$ & $corr \leq -0.14$ & $CV > 1.26$	Colon benign	0.08
	$STN > 0.73$ & $corr \leq -0.17$	Colon benign	0.06
	$corr \leq -0.17$ & $CV \leq 1.39$	Lung adenocarcinoma	0.12
	$corr > -0.07$ & $STN \leq 0.68$	Lung adenocarcinoma	0.07
	$corr > -0.16$	Lung benign	0.08
	$corr > -0.06$ & $STN \leq 0.69$	Lung benign	0.04
	$corr \leq -0.12$ & $CV \leq 1.26$	Lung squamous cell carcinoma	0.08
	$corr \leq 0.03$ & $CV > 1.30$ & $STN \leq 0.72$	Lung squamous cell carcinoma	0.04

of the key factors driving classification outcomes. The extracted rules highlight essential patterns and relationships in the data, supporting model validation and enabling a more transparent analysis of predictive behavior.

4.6. Statistical feature map overlay visualization explainability results

To evaluate the explainability of the proposed model and highlight the regions most influential to the model's predictions, the study employed the SFMOV technique, which overlays statistical features (mean, skewness, and entropy) onto the input images.

The results are assessed across five datasets, and for each dataset, the visualization for two representative images is presented. Each color intensity in the SFMOV is equivalent to the magnitude of the statistical measures, the warm colors (red) represents high values (more important regions) and cold colors (blue) represents low values (less important regions), providing a quantitative description of the attention areas of the DL model.

The resulting visualizations using SFMOV are analyzed and validated by a radiologist, confirming their clinical relevance and highlighting their potential for diagnostic decision-making. However, it is known that prior knowledge of the predicted class by the model can induce a reading bias in radiologists' reading of the overlays. Therefore, a blind reading process is advised, where the statistical overlays are read without looking at the predicted class first, in order to provide an unbiased assessment of the explainability of the model. This expert validation emphasizes the practical importance of the proposed visualization technique, particularly for medical imaging tasks requiring chest radiographs, Histopathological Images, MRIs, Ultrasound images, and retinal fundus images.

4.6.1. COVID-19 radiography database

The chest radiograph displays widespread haziness and bilateral lung opacities, particularly in the middle and lower zones. These findings are consistent with consolidation and ground-glass opacities, which are hallmarks of pneumonia associated with COVID-19.

Heatmap analysis of a COVID-19 image correctly classified as COVID-19 (Fig. 4, first row) provides valuable insights into the interpretability and decision-making of the model. Interestingly, the model appears to emphasize the relative absence of dense consolidations in the middle lung zones as a key feature for differentiation. The mean heatmap highlights inflammation and mild consolidation distributed across both lungs, focusing more on lower zones while avoiding significant central anomalies.

The skewness heatmap and entropy heatmap reveal patterns that suggest variability in peripheral and lower lung regions, but the middle zones remain less affected, aligning with radiological observations of COVID-19 in earlier stages or milder cases. The combined heatmap integrates these findings, emphasizing areas of bilateral involvement while leaving the middle zones relatively clean, which likely assists in distinguishing COVID-19 from other types of pneumonia.

This focus on the middle zones reflects the model's ability to recognize a critical radiological feature of COVID-19: the sparing of central

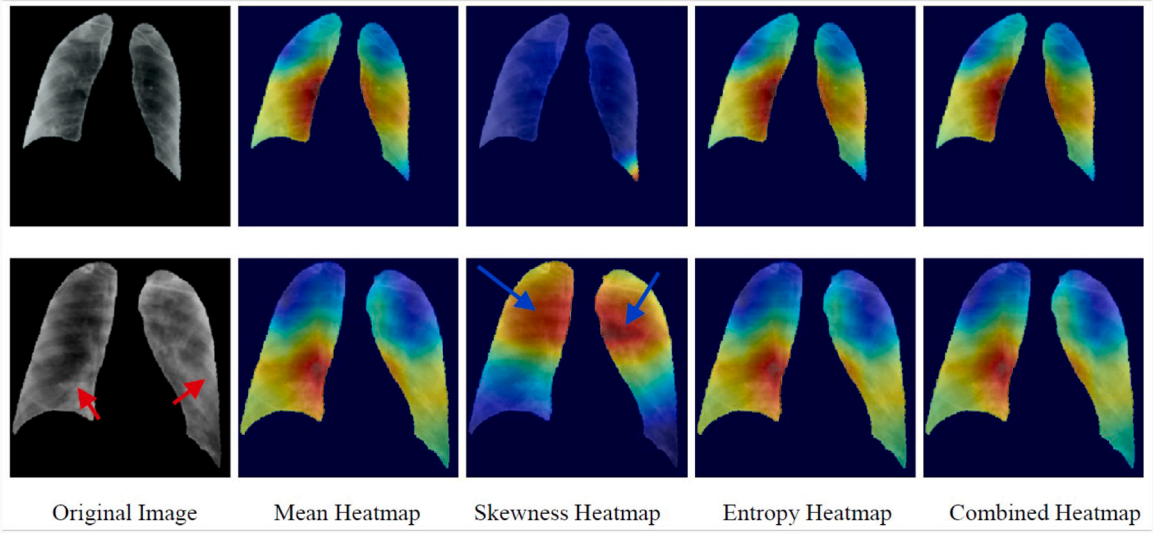


Fig. 4. SFMOV-based heatmap visualization showing a COVID-19 image detected as COVID-19 (row 1) and a COVID-19 image detection as viral pneumonia (row 2). Red arrows indicate consolidation and blue arrows indicate healthy parenchyma (the latter is what is focused on).

and middle lung regions from dense consolidations, a pattern often observed in viral pneumonias such as COVID-19, especially in contrast to bacterial pneumonias, which more frequently involve these regions.

In the case of a misclassification (COVID-19 labeled as viral pneumonia), the chest radiograph exhibits diffuse bilateral haziness and overlapping opacities consistent with both conditions (Fig. 4, second row). Here, the combined heatmap shows distributed attention without strong localization, likely due to a lack of distinguishing COVID-specific markers, such as clear sparing of the middle zones. This highlights the challenge of separating diseases with overlapping imaging features when the model relies heavily on specific radiological patterns.

The model appears to use the absence of dense consolidations in the middle zones as a key factor for correctly classifying COVID-19. While this approach aligns with known radiological patterns, it may fail in advanced cases of COVID-19 or in patients with mixed infections where central involvement is more prominent.

4.6.2. Ultrasound Breast Images for Breast Cancer

The analysis of heatmaps indicates that the model does not primarily focus on the core of the lesion in breast ultrasound imaging but rather on the interface between the lesion and the surrounding healthy tissue. This focus is particularly evident in the following cases:

- Malignant lesion detected as malignant (Fig. 5, Row 1). The heatmaps, especially the skewness heatmap, highlight the abrupt transitions at the interface between the lesion and healthy tissue, capturing irregular borders and textural asymmetries. While the mean heatmap partially highlights the lesion's core, it also emphasizes the boundaries and surrounding margins, suggesting that the model relies more on the interface for classification. The entropy heatmap reflects variability at the interface, rather than within the lesion itself, reinforcing the model's reliance on this transition zone. Clinically, this focus aligns with diagnostic practices, as irregular borders and abrupt changes in texture are key indicators of malignancy, often providing more diagnostic value than the lesion's internal features.

- Benign lesion detected as benign (Fig. 5, Row 2). For benign cases, the heatmaps highlight the smooth and homogeneous interface between the lesion and surrounding tissue. The mean heatmap and skewness heatmap emphasize the lesion's well-defined boundaries and smooth transitions into adjacent tissue, which are characteristic of benignity. The model pays minimal attention to the core of the lesion, instead relying on the uniformity of the margins and surrounding tissue.

4.6.3. Brain tumor "BTTypes"

The analysis of brain tumor MRI images demonstrates that the model focuses not only on the lesion itself but also on the surrounding tissue, particularly at the interface between the lesion and healthy tissue, which is crucial for differentiating malignancy. In the correctly classified malignant cases (Fig. 6, first row), the heatmaps highlight the lesion's irregular borders, heterogeneous internal structure, and variability in the immediate surroundings. This approach aligns with clinical expectations, as malignant tumors often influence adjacent tissue through invasion or edema, creating diagnostic patterns in these regions.

In benign cases correctly classified as benign (Fig. 6, second row), the model emphasizes characteristics such as smooth edges, well-defined margins, and homogeneous internal structures. The heatmaps reflect a focus on the peripheral areas of the lesion and the surrounding tissue, without highlighting irregular or heterogeneous patterns typically associated with malignancy. This behavior is consistent with clinical criteria for benign tumors, such as meningiomas, where smooth and uniform transitions are key diagnostic features.

These results indicate that the model relies on both the internal features of the lesion and its interaction with the surrounding tissue for decision-making. By focusing on relevant patterns for each case, the model demonstrates alignment with radiological criteria used by specialists, reinforcing its potential as a diagnostic tool for MRI-based brain tumor classification.

4.6.4. Lung and Colon Cancer Histopathological Images

In the case of colon adenocarcinoma image predicted as colon adenocarcinoma Fig. 7 (row 1), the original histopathological image shows tightly packed glandular structures with disorganized cellular morphology. Nuclear enlargement, hyperchromasia, and loss of polarity are evident, which are hallmark features of colon adenocarcinoma 7. These abnormalities stand out clearly in the stained tissue, making this an unmistakable case of malignancy.

Heatmap analysis confirms the model's focus on diagnostically significant regions. The mean heatmap highlights central areas with prominent nuclear irregularities and structural distortion, the skewness heatmap emphasizes structural variability along the areas of glandular disruption, and the entropy heatmap captures heterogeneity within the lesion. The combined heatmap (SFMOV) integrates these insights, focusing strongly on the regions of cellular and structural abnormalities. The high intensity in these areas confirms that the model is prioritizing the most diagnostically relevant features. Hence, the model's accurate

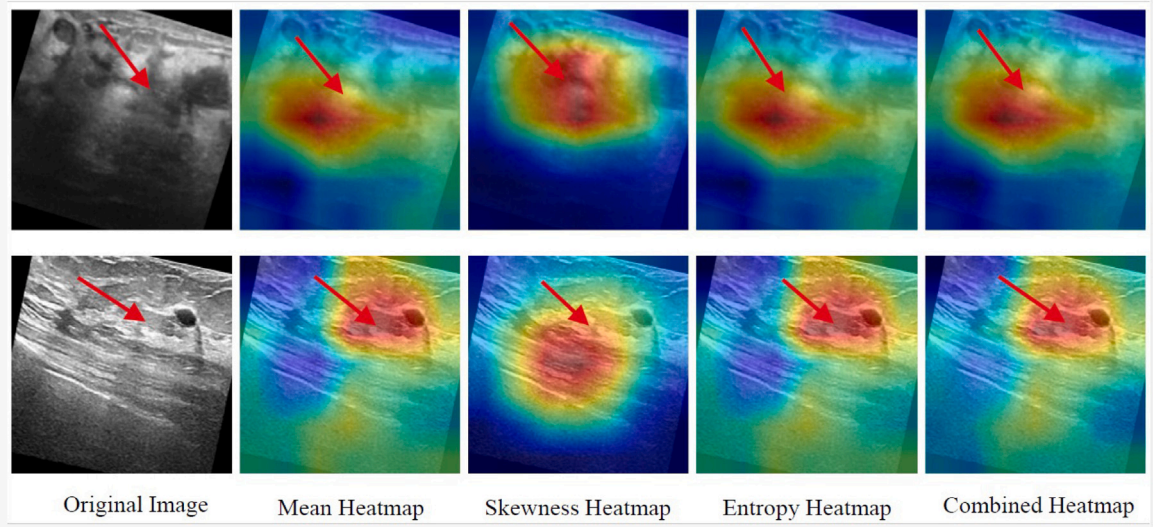


Fig. 5. SFMOV-based heatmap visualization showing a malignant ultrasound image detected as malignant (row 1) and a benign ultrasound image detection as benign (row 2). Red arrows represent the abrupt interface between healthy tissue and tumor.

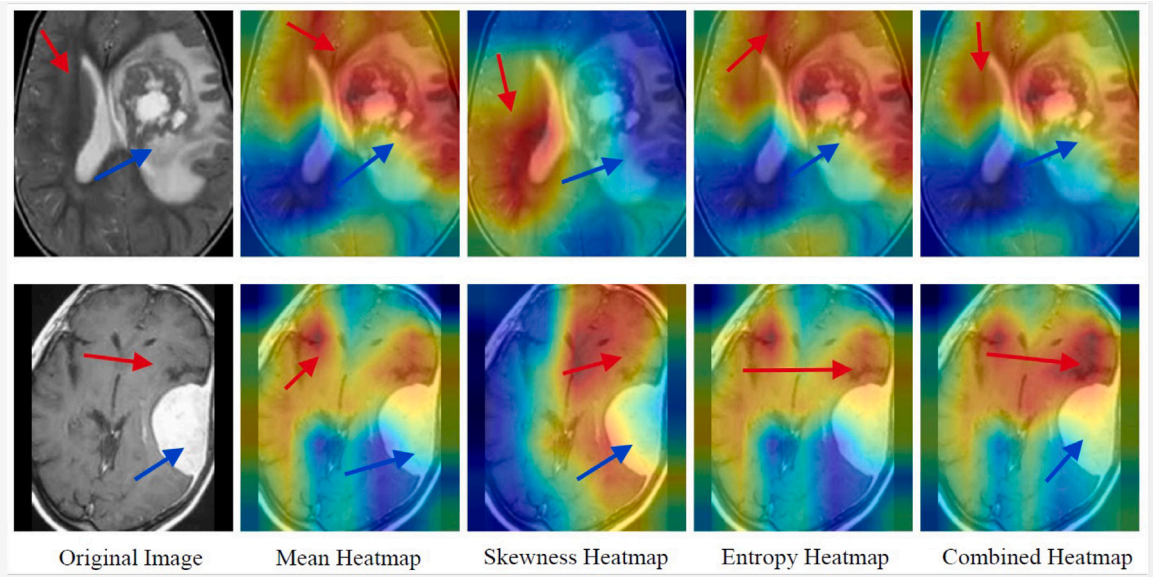


Fig. 6. SFMOV-based heatmap visualization showing a malignant image predicted as malignant (row 1) and a malignant image predicted as benign (row 2). Red arrows indicate the healthy tissue/healthy interface, while blue arrows indicate the tumor/lesion.

classification is supported by its emphasis on diagnostic features, such as disorganized glands and nuclear atypia, aligning with clinical criteria. Its precise attention to abnormal regions ensures consistent and reliable decision-making. By concentrating on key markers of malignancy with minimal distraction, the model demonstrates a logical and clinically relevant approach. This successful classification highlights the model's potential for reliable histopathological analysis, supporting its application in diagnostic workflows.

Furthermore, in the case of benign colon, predicted as benign colon Fig. 7 (second row), the histopathological image displays well-organized glandular structures with uniform cell morphology. The nuclei are regular in size, shape, and staining intensity, consistent with benign colonic tissue. No evident features of dysplasia or malignancy, such as nuclear pleomorphism or glandular disorganization, are observed. The mean heatmap emphasizes the central areas of the tissue, focusing on glandular structures, indicating the model's attention to normal tissue architecture. The skewness heatmap reflects low variability along the glandular regions, signifying homogeneity and lack of irregularity in the tissue. Similarly, the entropy heatmap highlights consistent and uniform patterns, aligning with the regular organization of benign colonic glands. The combined heatmap (SFMOV) integrates these metrics, showcasing a balanced focus on glandular regions and diagnostically relevant areas, further validating the benign classification. The classification is logical and well-supported by clinical criteria. The highlighted regions correspond to areas of uniform cell morphology and organized architecture, free from atypia or structural disruption, which are characteristics of benign colonic tissue. The model avoids focusing on non-diagnostic regions, such as the background, reflecting its ability to identify relevant features effectively. Low variability in the heatmaps, particularly in skewness and entropy, indicates high confidence in the classification.

It is worth noting that certain unintended visual patterns, such as the square-shaped high-intensity region observed in the bottom right corner of Dataset 4 images (Fig. 7) are observed. They are neither model-driven nor biological features but a result of the statistical

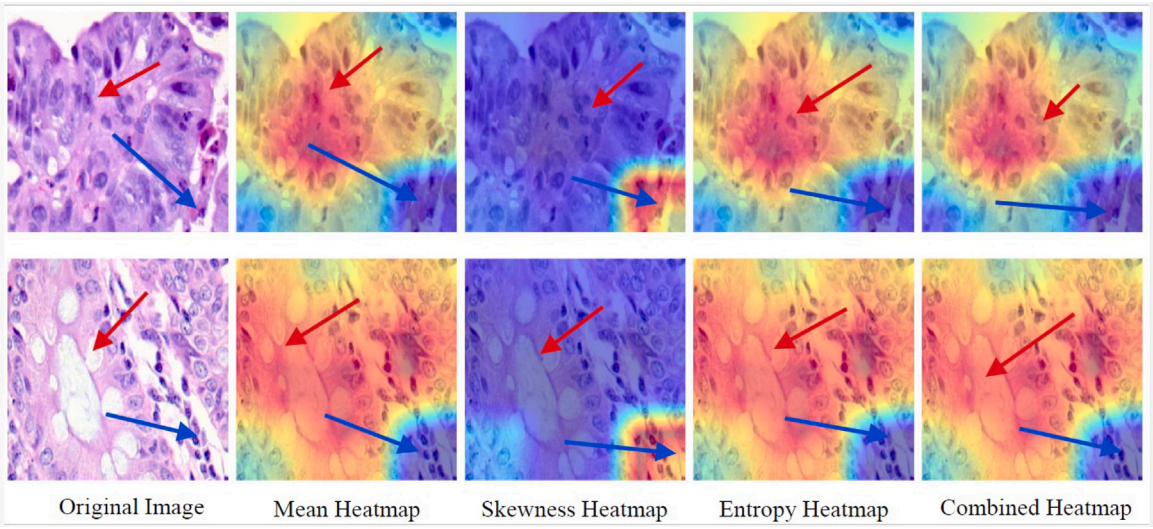


Fig. 7. SFMOV-based heatmap visualization showing a colon adenocarcinoma image predicted as colon adenocarcinoma (row 1) and a benign colon image predicted as benign colon (row 2). The pathological area is indicated by the blue arrow, whereas the red arrow indicates the healthy tissue/the transition zone.

feature computation process. That is, they are caused by local deformations of pixel intensity distributions when calculating skewness and entropy using a constant-sized kernel window. The heatmaps, including skewness and entropy, are generated by applying local statistical measures over small regions of the image. If a fixed size of the kernel is used, unexpected changes in intensity at a few locations may cause artificial structures that do not exist in any actual histopathological structures. This effect is a computational byproduct rather than an actual diagnostic feature and does not influence the overall model interpretation. Diagnostically relevant regions remain the primary target, and this ensures the model's classification remains clinically and biologically relevant.

4.6.5. ACRIMA from Generated Eye Dataset for Glaucoma Detection

The original retinal fundus images in the ACRIMA glaucoma detection dataset clearly show two morphological features, i.e., optic nerve head and vascular patterns, which are used by specialists to check for glaucomatous changes.

In the case of correct classification (glaucoma detected as glaucoma) as shown in the first row of Fig. 8, the focused attention and structural visibility of the optic nerve head are critical for evaluating potential glaucomatous damage and classification. The mean heatmap focuses strongly on the optic nerve head, meaning that the model uses this area in its prediction. The skewness heatmap confirms the existence of asymmetry in pixel values and intensity in specific locations around the optic disc, which corresponds to structural asymmetries such as cup-disc expansion that are frequently seen with glaucoma. The skewness heatmap confirms the presence of asymmetry in the pixel values and intensity in particular areas around the optic disc, which corresponds to structural asymmetries such as cup-disc enlargement, that are often observed with glaucoma. The entropy heatmap also focuses on areas around the optic nerve head, which is determined by fine textures and intensities of changes related to optic nerve head shifts. Integrating mean, skewness, and entropy heatmaps, the combined (SFMOV) heatmap also focuses on the area of the optic nerve head the papilla. As a result, to make decisions, the anatomical structures that are most pertinent to classifications must be present. The model's correct classification is logical because the model is trained to pay attention to regions like the optic nerve head and surrounding areas which are crucial in detecting glaucomatous changes. The attended aspects, including structural features and asymmetries, are consistent with clinical indicators in clinical practice like cupping, nerve fiber layer deflation, or vessel shift. Furthermore, the heatmaps demonstrate that the model is not distracted by other insignificant areas, indicating a sound decision-making method. This consistency with proven pathophysiological processes and clinical criteria, such as the optic cup-to-disc ratio or optic nerve atrophy, adds to the classification's accuracy and validity. Overall, by identifying the clinically important regions, the model succeeds in prediction, with the potential for comparable performance in future scenarios.

Furthermore, the heatmap analysis proves that the model correctly classified a typical retinal fundus image. The original image depicts a healthy optic nerve head with no glaucoma-related damage, such as cupping or nerve fiber weakening, as shown in the second row of Fig. 8. The mean heatmap emphasizes the optic nerve head and adjacent regions, concentrating on essential anatomical sites. The skewness heatmap reveals no notable asymmetries, which is compatible with the look of a healthy optic disc, and the entropy heatmap exhibits limited fluctuation, demonstrating the model's confidence. The combined (SFMOV) heatmap brings these data together, emphasizing the optic nerve head and verifying the absence of anomalies. The model's emphasis on the optic nerve head and adjacent areas is consistent with clinical guidelines, as these are critical for diagnosing glaucoma. The lack of anomalies, such as an elevated cup-to-disc ratio, supports the categorization. little entropy levels suggest strong confidence and little uncertainty. The heatmaps show that the model consistently prioritizes key characteristics while avoiding distractions. This strong match with clinical findings verifies the classification's accuracy and dependability.

4.7. Comparison of the proposed approach with non-interpretable models

In order to assess the performance of the proposed method, in this section, two popular non-interpretable models, ResNet50 and AlexNet, are compared with it on the same medical imaging datasets.

The classification results (accuracy, precision, recall, and F1-score) for the proposed method compared to different non-interpretable approaches on different data sets are presented in Table 14. The findings indicate that the suggested method performs better than ResNet50 and AlexNet on all the metrics considered for all the datasets. Although ResNet50 and AlexNet are strong models, they lack the same degree of interpretability as our suggested method. Our method incorporates statistical, visual, and rule-based explanations, making sure that the predictions of the model are not just precise but also interpretable, which is critical in clinical practice.

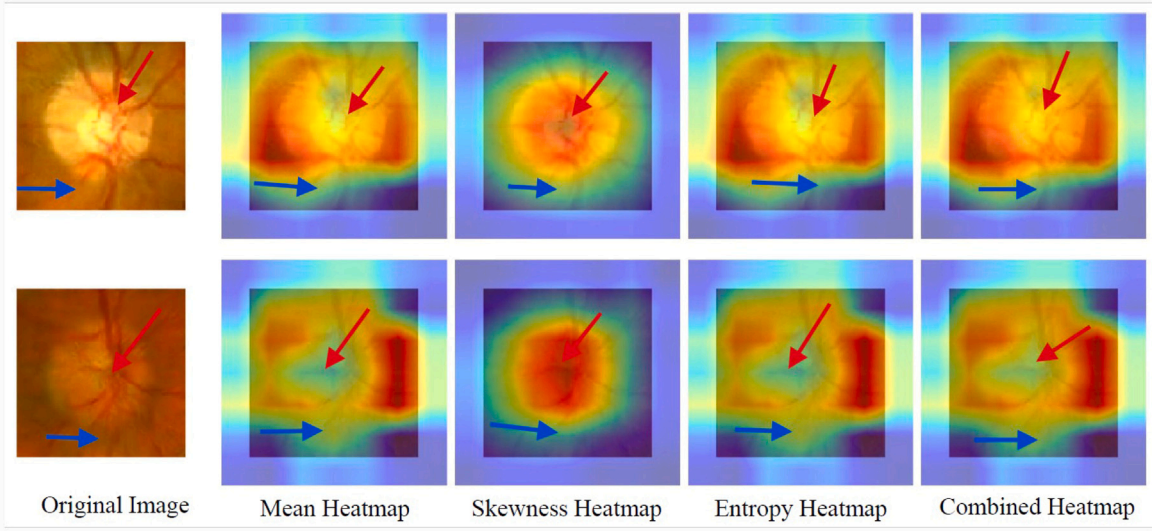


Fig. 8. SFMOV-based heatmap visualization showing a glaucoma image (row 1) and a normal image (row 2), both correctly classified, with model focus on clinically relevant optic nerve regions. Red arrows indicate the macular zone, while blue ones are the transition zone with the non-macular area.

Table 14
Classification performance of different noninterpretable approaches across multiple datasets.

Dataset	Model	Accuracy	Precision	Recall	F1-score
COVID-19 Radiography Database	ResNet50	70.75	69.69	70.75	68.69
	AlexNet	82.06	82.04	82.06	82.04
	Proposed approach	88.93	89.18	88.90	88.88
Ultrasound Breast Images for Breast Cancer	ResNet50	82.11	82.29	82.06	82.07
	AlexNet	83.30	83.31	83.30	83.30
	Proposed approach	85.29	85.16	85.71	85.21
BTTypes	ResNet50	89.00	89.00	89.00	89.00
	AlexNet	91.39	91.49	91.39	91.38
	Proposed approach	92.50	92.52	92.71	92.49
Lung and Colon Cancer Histopathological Images	ResNet50	72.37	72.47	72.37	71.54
	AlexNet	84.99	87.39	84.99	84.72
	Proposed approach	94.10	94.20	94.03	94.05
CRIMA from Generated Eye Dataset for Glaucoma	ResNet50	69.00	69.00	69.00	69.00
	AlexNet	80.44	81.24	80.44	80.32
	Proposed approach	84.56	84.53	84.64	84.54

This side-by-side comparison is intended to highlight the fact that our method successfully realizes a high performance-interpretability trade-off with both state-of-the-art predictive performance and interpretable, transparent decision-making. This is particularly crucial in the medical image analysis field, where model interpretability is directly related to user acceptance and trust.

4.8. Computational complexity and execution time of the proposed modules

This section presents the computational cost analysis of the proposed framework. Table 15 summarizes the computation times (in seconds) for each module across five medical imaging datasets.

The custom MobileNetV2 model is used because of its lightweight nature with 2,422,210 total parameters out of which 164,226 are trainable parameters. MobileNetV2's estimated floating point operations per second (FLOPs) are approximately 300 million, enabling faster training and inference times compared to heavier networks.

As can be seen in Table 15, training time for MobileNetV2 ranged from 511.43 s for the BTTypes dataset to 25,194.59 s for the Lung and Colon Cancer dataset. Deep feature extraction and statistical feature computation are computationally relatively fast too, with extraction times ranging from 169.20 to 1,033.28 s and statistical feature calculations ranging from 15.29 to 81.87 s.

The ZFMIS feature selection method is of negligible computational cost, requiring typically less than 1 s for all datasets. Similarly, DT requires less than 1 s for both training and testing on all datasets. RuleFit, though computationally costly, is within bounds, requiring between 79.41 and 468.07 s to train and less than 0.14 s to test.

For interpretability, the SFMOV process took between 4.34 and 7.66 s to create visual explanations for each image in real time. Additionally, hyperparameter tuning through grid search is done efficiently, with DT grid search taking between 9.48 and 37.69 s, and RuleFit grid search taking between 0.28 and 3,733.26 s, depending on dataset size and complexity. Furthermore, the Rule extraction stages of DT and RuleFit are also computationally effective, typically running in seconds.

These findings indicate that the proposed framework offers a reasonable trade-off between interpretability and computational efficiency, ensuring applicability to real-world deployment for medical image analysis tasks.

5. Conclusions

This study proposed SVIS-RULEX as an innovative framework to enhance the interpretability of deep learning models, particularly for critical applications in healthcare. By combining statistical feature engineering, rule-based modeling, and visual overlays, the methodology addresses key limitations of existing XAI approaches, offering a holistic solution for interpreting black-box models in medical diagnostics.

The framework demonstrated the ability to provide transparent, interpretable insights without compromising predictive performance.

Table 15
Computation time (in Seconds) for different modules on multiple datasets.

Module	BTTypes	ACRIMA from generated eye dataset for glaucoma detection	Ultrasound breast images for breast cancer	Chest radiography database	Lung and colon cancer histopathological images
MobileNet Training	511.43	3471.87	8190.62	20 534.81	25 194.59
Deep Feature Extraction	169.20	382.88	323.11	844.35	1033.28
Statistical Features Calculation	15.29	28.88	22.70	60.31	81.87
ZFMIS Feature Selection	0.26	0.37	0.33	0.68	0.80
DT Training Time on All Statistical Feature Set	0.06	0.19	0.16	0.53	0.66
DT Testing Time on All Statistical Feature Set	0.00	0.00	0.00	0.01	0.01
RuleFit Training Time on All Statistical Feature Set	468.07	118.55	79.41	159.23	183.51
RuleFit Testing Time on All Statistical Feature Set	0.01	0.06	0.03	0.10	0.14
SFMOV Time for One Image	7.66	6.50	4.50	4.34	4.94
Grid Search Time for DT	9.48	13.62	17.81	37.69	29.09
Grid Search Time for RuleFit	3733.26	1469.17	1006.19	0.28	2559.98
DT Rule Extraction	0.01	0.04	0.04	0.25	0.22
RuleFit Rule Extraction	1.33	2.67	2.22	16.98	13.71

By integrating statistical metrics such as mean, skewness, and entropy, the method enriched the interpretive capacity of deep learning models, while ZFMIS effectively reduced dimensionality, preserving only the most critical features. This approach facilitated the development of interpretable and high-performing models, offering valuable insights into the underlying patterns and decision-making processes of the model.

Visual overlays generated by SFMOV provided meaningful heatmaps, highlighting diagnostically relevant areas and demonstrating the model's focus on critical regions within medical images. These visualizations supported clinical trust in the decision-making process, ensuring that the results were interpretable and actionable.

The versatility of the framework was evident in its ability to perform consistently across a variety of medical imaging modalities, including radiographs, ultrasounds, and histopathological images. This scalability emphasizes its potential for broad application in diagnostics, making it particularly valuable in high-stakes domains like healthcare. Importantly, the rule-based explanations offered by decision trees and RuleFit models bridged the gap between computational predictions and human understanding, enabling a clearer interpretation of the model's decision-making process.

The study not only highlights the proposed framework's interpretability and transparency but also establishes its relevance for improving trust and adoption in real-world clinical settings.

Despite its promising contributions, the proposed SVIS-RULEX framework has certain limitations. While the rule-based explanations provide clarity, they might oversimplify the underlying decision-making process of deep learning models, potentially missing nuanced patterns that contribute to predictions. The reliance on visual heatmaps for interpretability also assumes that the highlighted regions fully capture diagnostically relevant features, which may not always align with domain experts' expectations. Furthermore, the framework's evaluation focused on specific imaging modalities and datasets, raising the need for extensive validation across broader, more heterogeneous datasets to ensure its generalizability.

It is worth noting that SFMOV does not necessarily coincide exactly with clinically marked tumor or consolidation areas. While these regions often overlap with areas typically examined by human experts during diagnosis, it is important to note that XAI methods, such as SFMOV, are not designed to exactly replicate expert attention. Instead, they may capture subtle, discriminative patterns that inform the model's decisions but do not always align with clinically defined tumor or consolidation boundaries.

The current SFMOV approach employs statistical feature maps (mean, skewness, entropy), computed directly from the original activation maps of the final convolutional layer without applying additional learned or domain-specific weights. This deliberate choice preserves the direct interpretability of the features of the learned model and avoids introducing external bias, ensuring that the visualizations remain faithful to the representations of the inherent features of the network. Although this method is computationally efficient and effectively highlights regions of model attention, the absence of learned weighting may result in broader, less localized feature activations. However, integrating the learned weights could improve visualization clarity while preserving the integrity of the model.

Furthermore, another limitation of the proposed method is that, for some data sets, the SFMOV combined heatmaps are visually very close to the mean heatmap. This closeness means that the effect of the mean feature can dominate the visualization and could undermine the contribution of skewness and entropy under such scenarios. Future work will focus on addressing this limitation by enhancing the weighting of statistical features in a way that makes the discrimination higher in combined heatmaps.

Finally, the computational overhead introduced by combining statistical, rule-based, and visual methods may pose challenges for real-time deployment in resource-constrained settings. Addressing these limitations in future work will be essential to further enhance the framework's robustness and applicability in diverse and practical clinical environments.

CRediT authorship contribution statement

Naeem Ullah: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Florentina Guzmán-Aroca:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Francisco Martínez-Álvarez:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Ivanoe De Falco:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Giovanna Sannino:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Code availability

The source code for the proposed method is publicly available at the following GitHub repository: https://github.com/Engr-Naeem-Ullah/XAI-Med-Images-Stat-Visual-Rules. This facilitates reproducibility, enables validation of the results, and encourages further research and development in the field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data used in the experiments are freely available. In the paper, all databases have been referenced.

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